

The Search Costs of Inflation in the Labor Market^{*}

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Abstract

This paper studies the effect of unanticipated inflation in the labor market. When wages are contracted in nominal terms, inflation reduces real wages, leading workers to intensify on-the-job search to obtain wage-adjusting outside offers. Both the search effort and the resulting job mobility are costly, yet that same mobility raises output by moving workers toward more productive matches. To quantify these costs and benefits, we extend the canonical job ladder of Postel-Vinay and Robin (2002) to a nominal environment with privately chosen search effort, and calibrate it to standard moments from the pre-pandemic US labor market. A one-time inflation shock scaled to the COVID episode generates an average welfare loss of 0.44% of consumption, with workers at the top of the wage distribution bearing losses over six times those at the bottom. Once the offsetting transfer of surplus to firms and productivity gains from reallocation are accounted for, the net aggregate cost is close to zero. Second, inflation volatility imposes asymmetric costs: worker welfare is approximately invariant to the volatility regime because firms compensate workers for expected search costs at hiring, but social welfare declines meaningfully because the resource cost of search effort is not undone by compensation. This second finding identifies a distinct, labor-market source of welfare costs from inflation volatility, complementing the price-dispersion costs emphasized in the New Keynesian literature.

JEL Codes: J3, J6

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1 Introduction

When asked directly why they dislike inflation, households most commonly cite concerns that it “diminishes their buying power, as neither personal nor general wage increases seem to match the pace of rising prices” (Stantcheva (2024)), and that it erodes their standard of living (Shiller (1997)). These claims suggest that households may search and change jobs to protect their purchasing power from inflation. Recent evidence supports this intuition. “The Great Resignation” of 2021–2022, during which quit rates and job-to-job transition rates reached historic highs alongside the post-pandemic inflation surge, raised the question of whether elevated inflation might be driving workers’ search behavior. Indeed, Pilossoph and Ryngaert (2025) find that when facing nominally rigid wages, workers respond to elevated inflation expectations by searching harder while employed, a natural response given that on-the-job search is a primary mechanism through which workers achieve wage growth over time (Topel and Ward (1992)). Together, these household perceptions and behavioral responses point to an underexplored channel through which inflation affects welfare: the search and mobility decisions of workers in the labor market. On the one hand, inflation-induced search is costly, requiring workers to expend effort and incur mobility costs. On the other, since job-to-job transitions are closely linked to productivity growth (Moscarini and Postel-Vinay (2018)), inflation can deliver benefits by moving workers up the job ladder. This paper develops a theory of these costs and benefits and provides estimates of their magnitudes.

To this end, we extend the canonical job ladder of Postel-Vinay and Robin (2002) along two dimensions. First, wages are contracted in nominal terms: contracts are indexed to trend inflation, analogous to a cost-of-living adjustment, but nominal wages are not indexed to inflation shocks; these shocks therefore erode real wages unless renegotiation occurs. Second, search effort is endogenous and privately chosen by workers.¹ Because of two-sided lack of commitment, nominal wages of incumbent workers can only adjust beyond contractual indexation through credible outside offers. Inflation shocks therefore erode real wages for incumbents until such an offer arrives, transferring surplus to employers. Workers respond by intensifying search to raise the probability of receiving an outside offer, which they use either to renegotiate their current wage or to switch jobs. Both search and job switching are costly, creating a tradeoff between the costs of action (search and mobility) and the costs of inaction (real wage losses).

The theory has two key implications. First, and in line with the basic narrative from surveys, workers will dislike (unexpected) inflation since it either comes with lower real wages, or requires costly search and mobility to avoid it. Second, and as a direct consequence of the rise in search effort that ensues to mitigate its impact, inflation shocks have allocative consequences. As search effort increases, workers can not only increase their wages while remaining with their employer,

¹Faberman et al. (2022) introduced endogenous search into this class of (real) models under the alternative assumption that effort is bilaterally chosen with the firm.

but also move up the job ladder to more productive firms. Together, these forces determine the aggregate welfare effects of unexpected inflationary shocks, which theoretically may be positive or negative. Using our quantitative model, we find that the costs borne by workers are offset via gains that accrue to firms.

These allocative effects raise a question of efficiency. Because workers choose search effort to maximize their own value, their response to inflation is not generally efficient. A planner who sets search directly would weigh only the cost of search against the expected productivity gains from reallocation, and would therefore leave search unchanged when inflation shifts surplus between a worker and her employer. Workers internalize neither force fully, and the direction of the resulting distortion depends on the workers' position on the ladder. Near the bottom, where the uninternalized gains from reallocation are large, workers search too little; near the top, where a successful search yields no reallocation and simply bids up the wage at the current employer, workers expend too much effort over a transfer with no social value. Inflation shocks exacerbate exactly this margin: by cutting real wages, they raise search effort everywhere, bringing some workers further from what the planner would like and some workers closer. In sum, the search workers undertake to recover lost wages is partly productive reallocation, which is socially valuable, and partly a contest over the division of surplus, which is a pure resource cost. It is the second component that may make inflation costly in the aggregate even when workers are compensated for it.

We calibrate the model to match key features of the pre-pandemic US labor market, including the real wage growth of stayers and movers, the search-wage elasticity, and the job-offer yield (the probability an offer converts to a job) for the employed and non-employed. We then use the calibrated model in two quantitative exercises. The first considers a one-time, unanticipated inflation shock scaled to the COVID episode, in which inflation rises from 2% to 9% on impact and returns slowly to trend. The average worker welfare loss is 0.44% in consumption-equivalent terms, but this average conceals substantial heterogeneity. Workers at the top of the wage distribution absorb the real wage losses and would give up roughly 0.9% of flow consumption to avoid the shock. Workers with high wages are largely those already at the top of the job ladder, so search does little to raise their wages: few outside firms are more productive, and thus few can credibly threaten to bid their wages up. The costs borne by workers at the bottom of the distribution are one-sixth of those at the top: rather than absorbing a permanent real wage loss, these workers pay search and mobility costs to climb the ladder and eventually recover. Once we account for the offsetting transfer of surplus to firms and the productivity gains from reallocation, the net social cost of the shock falls to essentially zero, suggesting that productivity improvements offset any search and mobility costs that workers incur. The second exercise studies the welfare consequences of ongoing inflation volatility around trend. We find that worker welfare is approximately invariant to the volatility regime, since firms compensate

workers for expected search costs through higher initial wage levels. Aggregate welfare, however, declines steeply with inflation volatility. At US volatility levels, the social loss is small, but at levels observed in many emerging-market economies, it becomes economically meaningful.

Our paper is closest in spirit to contemporaneous work by Afrouzi et al. (2024) and Guerreiro et al. (2025), both of which develop theories of why workers dislike inflation.² The three papers share two methodological premises. First, inflation is taken as exogenous in order to study its costs rather than its causes. Second, the friction that generates those costs operates through nominal wage rigidity. Both premises distinguish this work from a related literature in which inflation arises endogenously from distributional conflict between workers and firms over real wages (Lorenzoni and Werning (2023)). The three papers differ along two further dimensions. The first is which friction gives rise to the costs. Guerreiro et al. (2025) build a sticky-wage framework in which renegotiation entails an explicit conflict cost, paid each time wages adjust. Afrouzi et al. (2024) combine search costs with renegotiation costs in a related setting. In our framework renegotiation is not directly costly: it is the consequence of costly search, and the link between search and wage growth through renegotiation is disciplined with standard moments, including job-to-job transition rates, wage growth for movers and stayers, and the search-wage elasticity. The second difference is the welfare margin our framework can quantify. The job-ladder structure of our model allows us to measure the productivity gains from inflation-induced search and reallocation, since workers who intensify search are more likely to reallocate to more productive matches. We find these gains are substantial: the net social cost of a COVID-sized inflation shock falls from 0.1% of consumption to effectively zero once these gains are accounted for.

The job ladder also connects our paper to the literature on inflation and reallocation. The goods-pricing side of our model is exogenous and therefore far less rich compared to Moscarini and Postel-Vinay (2023), but we add an endogenous response of the offer arrival rate to inflation through search effort, with inflation shocks raising both job-to-job transitions and within-firm renegotiations simultaneously. This is suggestive of a wage-price spiral mechanism (Blanchard (1986), Lorenzoni and Werning (2023)): Faccini and Melosi (2025) and Karahan et al. (2017) document that on-the-job search and job-to-job transition rates predict future inflation. Our results show how that empirical pattern can emerge endogenously when search effort responds to inflation.

Our modeling choice of nominal contracts with limited indexation in a Bertrand-competition environment sits within a broader literature on nominal rigidity in search models. Gertler et al. (2008) and related work introduce staggered nominal Nash bargaining into the Mortensen-Pissarides framework. A separate strand, including Gertler and Trigari (2009) and Gertler et al. (2020), models staggered bargaining over real wages; in those frameworks, inflation surprises are

²See also Blanco et al. (2026) for a general theory of inefficient quits and inflation.

by construction absorbed into wage levels and do not erode purchasing power, so the channel we study does not operate. More recently, Graves et al. (2023) document that worker-side margins (e.g. quitting and searching) are an active component in the transmission of monetary policy shocks, consistent with the mechanism in our model. Relative to all of this work, our contribution is to bring the nominal-contracting friction into a job-ladder framework rich enough to discipline the search response with standard labor market moments and to quantify the resulting welfare costs.

A related literature studies the distributional incidence of inflation through its effects on the value of nominal assets (Doepke and Schneider (2006a); Del Canto et al. (2025)). Our analysis is complementary: we abstract from asset-side channels and focus on the labor market as a distinct avenue through which inflation imposes welfare costs. An important distinction between the two channels, however, is that the labor-market channel features both a transfer component (between workers and firms) and a resource-cost component. A decline in real wages induced by an inflationary shock redistributes surplus from workers to firms, much like the nominal-wealth redistribution in the asset-price literature, but the search activity workers undertake to mitigate that decline is a real resource cost that no compensating transfer can undo.

Finally, this paper contributes to the literature on the passthrough of price inflation to wage inflation. Hajdini et al. (2022), Jain et al. (2024), and Buchheim et al. (2024) document that the perceived passthrough of inflation to wage growth is low. Pilossoph and Ryngaert (2025) show this low perceived passthrough prompts employed workers to intensify search, potentially accelerating the arrival of wage-adjusting offers.³ Our model combines nominal wage rigidity and endogenous private search to study how realized inflation passes through to wage growth in equilibrium, allowing for wage growth through both job mobility and renegotiations on-the-job.

The paper proceeds as follows. Section 2 develops the model and discusses its efficiency properties. Section 3 estimates the model and discusses the model fit vis-à-vis the data. Section 4 performs two counterfactuals aimed at quantifying the costs of inflation in the labor market. The first is a one-time, unanticipated shock to inflation meant to mimic the COVID-19 episode; the second is a comparison of our baseline economy to an economy with no shocks to inflation around trend. Section 5 concludes.

2 Model

The model extends the canonical job ladder of Postel-Vinay and Robin (2002) in two dimensions. First, wage contracts are written in nominal terms and indexed to trend inflation; deviations

³Buchheim et al. (2024) find that among German workers, expected passthrough rises when workers and firms anticipate negotiations, but higher inflation expectations do not increase workers' propensity to enter negotiations. Pilossoph and Ryngaert (2025), however, do find evidence of increases in the propensity to ask for a raise in U.S. data in their higher inflation hypothetical.

of realized inflation from trend are not indexed in the contract, so the real-wage erosion they cause can only be undone through the same renegotiation channel that governs wage growth in the baseline framework. Second, search effort is endogenous and privately chosen by workers, following Faberman et al. (2022), who introduce endogenous search into this class of models under the alternative assumption that effort is bilaterally chosen with the firm. All other features of the environment remain standard. The combination of these two ingredients is what allows the model to reproduce the central empirical finding in Pilossoph and Ryngaert (2025): workers facing nominally rigid wages intensify on-the-job search in response to higher expected inflation. Aggregate productivity z and the aggregate price level P are exogenous; we describe each in turn.

Productivity grows deterministically at a constant rate, $g_z > 0$. The exogenous price level consists of two components: a deterministic trend component p and a stochastic component ε , $P = p \cdot \varepsilon$. The deterministic trend component grows at constant rate $g_p > 0$:

$$p' = p(1 + g_p)$$

The stochastic price level component grows at a rate determined by ε' :

$$\varepsilon' = \varepsilon(1 + g'_\varepsilon)$$

where the shock g'_ε follows:

$$g'_\varepsilon = \rho g_\varepsilon + \nu', \quad \nu' \sim N(0, \sigma_\nu^2)$$

There is a unit mass of firms, each with a vacancy, indexed by their productivity $y \in (\underline{y}, \bar{y})$. The exogenous distribution of vacancies across firms y is denoted by $F(y)$.⁴ Per-period real output between a worker and a firm depends on aggregate productivity and the firm type, y , such that $Y(z, y) = z \cdot y$.

Workers are homogeneous, infinitely lived, and of measure one. They have linear preferences over a single final consumption good. Workers can be employed or unemployed, and we denote those states by $i \in \{e, u\}$, respectively. Both employed and unemployed workers make search effort decisions $s \in (0, \bar{s}_i)$ which determine the rate at which they meet available vacancies. For employed workers, the rate is $\lambda_e + s$ and for the unemployed it is $\lambda_u + s$.⁵ The real cost of search is given by $C(z, s)$, which is increasing and convex in search effort, $\frac{\partial C(z, s)}{\partial s} > 0$,

⁴It is theoretically straightforward to introduce endogenous vacancy creation so that there is a notion of tightness which moves with inflation. However, with endogenous search effort, the vacancy creation decision becomes a function of the distribution of workers across firms and the unemployment rate, which substantially increases the computational burden of the model. Instead, Appendix F introduces a correlation between the exogenous arrival rate of offers and the cyclical component of inflation in order to account for the co-movement of unemployment and inflation over the cycle.

⁵The upper bound on search is therefore $\bar{s}_e = 1 - \lambda_e$ for the employed and $\bar{s}_u = 1 - \lambda_u$ for the unemployed.

$\frac{\partial^2 C(z,s)}{\partial^2 s} > 0$. Workers exogenously separate at rate δ from their jobs, and earn a real flow value of unemployment $B(z)$. All agents discount the future at rate β .

In what follows, we denote by $U(\cdot)$, $W(\cdot)$ and $J(\cdot)$ the value of unemployment, the value of employment to the worker, and the value of employment to the firm, respectively; we derive these objects below.

2.1 Wage Setting and Wage Contracts

When firms meet unemployed workers, they make take-it-or-leave-it (TIOLI) offers, setting the initial nominal wage $w = \phi_u(y, p', \varepsilon', g'_\varepsilon, z')$ to leave the worker indifferent between employment and unemployment. The value of unemployment to a worker can thus be written as:

$$\begin{aligned} U(z) &= \max_{s \in [0, \bar{s}_u]} B(z) - C(z, s) + \beta(1 - (s + \lambda_u)) U(z') \\ &\quad + \beta \mathbf{E}_{g'_\varepsilon} (\lambda_u + s) \int_y \max \left\{ U(z'), W(\phi_u(y, p', \varepsilon', g'_\varepsilon, z'), y, p', \varepsilon', g'_\varepsilon, z') \right\} dF(y) \\ &= B(z) + \beta U(z') \end{aligned}$$

The second equality comes as a direct result of the TIOLI offers assumption: firms will offer workers a wage that is just large enough so that they are indifferent between taking the offer and remaining unemployed so that the future value collapses to a simple expression. Because workers get the same value whether they search or not, and search is costly, all unemployed workers choose $s_u^* = 0$.⁶

When a worker and firm form a match, they agree that the initial nominal wage w will grow deterministically at rate $(1 + g_z)(1 + g_p)$, tracking trend productivity growth and trend inflation. The wage level is renegotiated only by mutual consent following a triggering event: an outside offer, or a realization of the stochastic price-level component that pushes the real wage outside the range over which the match remains viable.

Absent an outside offer, a shock to the stochastic price level component g'_ε may render the real wage too low (high), which will cause the nominal wage to be adjusted upward (downward). We denote the adjusted wage *before an outside offer arrives* as w' , which is determined following:

$$w' = \begin{cases} \hat{w} : J(\hat{w}, y, p', \varepsilon', g'_\varepsilon, z') = 0, & \text{if } J(w(1 + g_z)(1 + g_p), y, p', \varepsilon', g'_\varepsilon, z') < 0 \\ \hat{w} : W(\hat{w}, y, p', \varepsilon', g'_\varepsilon, z') - U(z') = 0, & \text{if } W(w(1 + g_z)(1 + g_p), y, p', \varepsilon', g'_\varepsilon, z') - U(z') < 0, \\ w(1 + g_z)(1 + g_p), & \text{else} \end{cases} \quad (\text{LOM1})$$

⁶While this is not an accurate description of the search behavior of unemployed workers, our focus is on the

Equation LOM1 governs the within-match nominal wage adjustment before any outside contacts are made. Three cases arise. If the cyclical inflation shock is large and negative — so the real wage rises enough that the firm would prefer separation at the contracted nominal wage — the nominal wage is cut to leave the firm indifferent between retaining and releasing the worker. If the shock is large and positive — so the real wage falls enough that the worker would prefer unemployment — the nominal wage is raised preemptively to leave the worker indifferent between staying and quitting into unemployment. In all other cases the nominal wage grows according to the contract.⁷ The second case is the one relevant for a positive inflation shock and underlies the equilibrium response we study below: firms adjust nominal wages upward when necessary to avoid quits into non-employment, even before any outside offer arrives.

Wages may be renegotiated beyond w' if a worker receives an outside offer which dominates what they currently make, forcing the firm to adjust the base pay upward or allow the worker to leave for the new firm. The new wage, either on the job or at the new job, is determined by Bertrand competition between the incumbent firm and the poaching firm (Postel-Vinay and Robin (2002)). Consider a worker currently with a firm of type y_1 and making nominal wage w' . Suppose the worker is contacted by a firm of type y_2 . Bertrand competition between employers will lead to one of the three following outcomes:

Case 1: No outbidding, wage remains the same. The worker will not be able to use the outside offer to change their current wage if $y_2 < y_1$ and

$$W\left(\hat{w}(y_2, p', \varepsilon', g'_\varepsilon, z'), y_2, p', \varepsilon', g'_\varepsilon, z'\right) \leq W\left(w', y_1, p', \varepsilon', g'_\varepsilon, z'\right) \quad (1)$$

where $\hat{w}(\cdot)$ is defined implicitly as the maximum wage firm y_2 can offer the worker such that:⁸

$$J(\hat{w}, y_2, p', \varepsilon', g'_\varepsilon, z') = 0$$

The left-hand-side of Equation 1 is the value of employment to the worker employed at y_2 earning the maximum nominal wage firm y_2 is willing to pay when the price level is $p' \cdot \varepsilon'$ and aggregate productivity is z' . The right-hand-side is the value of employment to the worker of remaining at firm y_1 in the same realization of states and earning the nominal wage w' . Because the maximum nominal wage the poaching firm can offer to start is $\hat{w}(\cdot)$ and this provides a

search behavior of the employed, so we make this simplifying assumption. The assumption can easily be relaxed by either allowing workers to receive some value in employment that is a share of the value of unemployment (Cahuc et al. (2006)), or by making the value of unemployment depend on the price level through a benefit which is given nominally. In both cases, searching would have a positive return for the unemployed.

⁷A stronger form of downward nominal wage rigidity (DNWR) would replace case (i) with a separation: the firm would be unwilling to retain the worker at the unchanged nominal wage, but a downward adjustment would not be permitted. Our specification preserves the match in this region, which makes the model conservative with respect to inflation's effects. DNWR would amplify the costs we document.

⁸Without persistence in the cyclical inflation shocks, this wage would equal the full nominal output $y_2 z' p' \varepsilon'$.

value that is lower than the value the worker receives by remaining at the current firm at wage w' , the worker has no credible threat, and the contract remains unchanged.

Case 2: No outbidding, wage contract is renegotiated to ϕ^{reneg} . Suppose that $y_2 < y_1$, but instead:

$$W\left(\hat{w}(y_2, p', \varepsilon', g'_\varepsilon, z'), y_2, p', \varepsilon', g'_\varepsilon, z'\right) > W\left(w', y_1, p', \varepsilon', g'_\varepsilon, z'\right) \quad (2)$$

In this case, the worker indeed has a credible threat as the poaching firm is able to offer $\hat{w}(y_2, p', \varepsilon', g'_\varepsilon, z')$ and make the worker better off. The incumbent firm then must offer a wage, call it ϕ^{reneg} , which will make the worker indifferent between staying and leaving:

$$W\left(\hat{w}(y_2, p', \varepsilon', g'_\varepsilon, z'), y_2, p', \varepsilon', g'_\varepsilon, z'\right) = W\left(\phi^{\text{reneg}}, y_1, p', \varepsilon', g'_\varepsilon, z'\right)$$

Case 3: Poaching firm hires worker at wage ϕ^{poach} . If $y_2 > y_1$, the worker moves to the poaching firm, and gets a new wage ϕ^{poach} which satisfies:

$$W\left(\phi^{\text{poach}}, y_2, p', \varepsilon', g'_\varepsilon, z'\right) = W\left(\hat{w}(y_1, p', \varepsilon', g'_\varepsilon, z'), y_1, p', \varepsilon', g'_\varepsilon, z'\right)$$

The cases above imply that there is a cutoff firm, call it $q(w', y_1, p', \varepsilon', g'_\varepsilon, z')$, which is the lowest productivity firm that triggers a wage renegotiation or a job-to-job transition for a worker at firm y_1 employed at wage w' . This firm is defined implicitly as the firm q such that:

$$W(w', y_1, p', \varepsilon', g'_\varepsilon, z') = W(\hat{w}(q, p', \varepsilon', g'_\varepsilon, z'), q, p', \varepsilon', g'_\varepsilon, z') \quad (3)$$

If a worker receives an offer from firm $y_2 < q(w', y_1, p', \varepsilon', g'_\varepsilon, z')$, nothing happens. Above this value, but below the current firm y_1 , a wage renegotiation is triggered without a job-to-job transition. Above y_1 , the worker moves to the new firm. Figure 1 depicts these regions graphically.

Importantly, this cut-off firm depends not only on the firm at which the worker is employed, but also on her current nominal wage.⁹ For a given price level, the lower the current nominal wage, the lower is the cutoff firm, implying that a larger set of firms will induce a wage renegotiation if the worker receives an outside offer. Moreover, for a given nominal wage, a higher stochastic price level component ε (corresponding to higher realized inflation g_ε) lowers the real wage. This also lowers the cutoff firm $q(\cdot)$, producing the same effect.

The cutoff $q(\cdot)$ is the central object for understanding how price-level shocks transmit to the labor market. Holding the nominal wage w' fixed, a higher realization of ε' reduces the

⁹The latter is the case because search effort is chosen by the worker to maximize her own value rather than chosen bilaterally by both the firm and worker to maximize match value. We return to this point below.

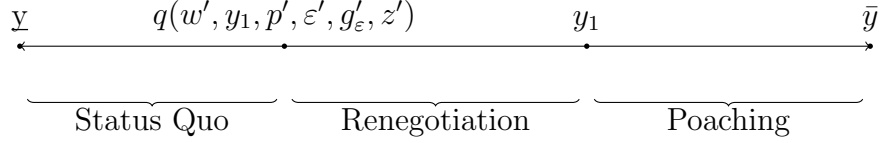


Figure 1: Offers Ranges for Status Quo, Renegotiation, and Poaching.

worker's real wage and, as shown formally below, reduces the value of employment at the current match. To preserve the indifference condition (Equation 3), $q(\cdot)$ must fall — the set of poaching firms that constitute a credible threat expands. This expansion has two consequences. First, conditional on receiving an outside offer, the worker is more likely to obtain a wage increase, either through renegotiation or a job-to-job move, than if their real wage had not declined. Second, because the expected return to receiving an outside offer rises, the worker chooses higher search effort. Both forces operate together: inflationary shocks raise job-to-job transitions, renegotiation rates, and search effort simultaneously. The search-effort response is the model counterpart to the empirical finding in Pilossoph and Rynngaert (2025).

With these cases in mind, we can now write the value of employment to a worker with current nominal wage w employed at firm y when the trend component of the price level is p , the stochastic component is ε , the cyclical component is g_ε , and productivity is z .

$$\begin{aligned}
W(w, y, p, \varepsilon, g_\varepsilon, z) &= \max_{s \in [0, \bar{s}_e]} \frac{w}{p \cdot \varepsilon} - C(z, s) + \beta \delta U(z') \\
&+ \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta) (s + \lambda_e) \int_{\underline{y}}^{q(w', y, p', \varepsilon', g'_\varepsilon, z')} W(w', y, p', \varepsilon', g'_\varepsilon, z') dF(x) \\
&+ \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta) (s + \lambda_e) \int_{q(w', y, p', \varepsilon', g'_\varepsilon, z')}^y W(\phi^{\text{reneg}}(y, x, p', \varepsilon', g'_\varepsilon, z'), y, p', \varepsilon', g'_\varepsilon, z') dF(x) \\
&+ \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta) (s + \lambda_e) \int_y^{\bar{y}} W(\phi^{\text{poach}}(x, y, p', \varepsilon', g'_\varepsilon, z'), x, p', \varepsilon', g'_\varepsilon, z') dF(x) \\
&+ \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta) (1 - s - \lambda_e) W(w', y, p', \varepsilon', g'_\varepsilon, z')
\end{aligned}$$

where

$$p' = p(1 + g_p) \quad (\text{LOM2})$$

$$\varepsilon' = \varepsilon(1 + g'_\varepsilon) \quad (\text{LOM3})$$

$$z' = z(1 + g_z) \quad (\text{LOM4})$$

and w' satisfies Equation LOM1. The worker earns a real wage of $\frac{w}{p \cdot \varepsilon}$, and pays a search cost $C(z, s)$ given some chosen effort s . With probability δ she separates into unemployment. With complementary probability, she remains employed. If she receives an outside offer, which occurs with probability $s + \lambda_e$, one of the three cases outlined above will occur.

Specifically, she will either (i) receive an offer from a firm x which does not constitute a credible threat ($x < q(w', y, p', \varepsilon', g'_\varepsilon, z')$), and her nominal wage will adjust according to w' described above, (ii) receive an offer from a firm x which induces a change in the contracted wage to $\phi^{\text{reneg}}(y, x, p', \varepsilon', g'_\varepsilon, z')$, but no job-to-job transition will occur ($q(w', y, p', \varepsilon', g'_\varepsilon, z') \leq x \leq y$), or (iii) receive an offer from a firm which induces a job-to-job transition ($x > y$), where she receives a wage $\phi^{\text{poach}}(x, y, p', \varepsilon', g'_\varepsilon, z')$. Finally, the worker may not receive an offer, in which case the wage adjusts to w' .

Optimal search effort for this worker will satisfy:

$$s_e^*(w, y, p, \varepsilon, g_\varepsilon, z) = \begin{cases} 0, & \text{if } \tilde{s}_e(w, y, p, \varepsilon, g_\varepsilon, z) < 0, \\ \tilde{s}_e(w, y, p, \varepsilon, g_\varepsilon, z), & \text{if } 0 \leq \tilde{s}_e(w, y, p, \varepsilon, g_\varepsilon, z) \leq \bar{s}_e, \\ \bar{s}_e, & \text{if } \tilde{s}_e(w, y, p, \varepsilon, g_\varepsilon, z) > \bar{s}_e \end{cases} \quad (\text{OS1})$$

where $\tilde{s}_e(\cdot)$ is defined as:

$$\begin{aligned} \frac{\partial C(z, s)}{\partial s} \Big|_{\tilde{s}_e(w, y, p, \varepsilon, g_\varepsilon, z)} = & \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta) \int_{q(w', y, p', \varepsilon', g'_\varepsilon, z')}^y \left(W(\phi^{\text{reneg}}, y, p', \varepsilon', g'_\varepsilon, z') - W(w', y, p', \varepsilon', g'_\varepsilon, z') \right) dF(x) \quad (\text{OS2}) \\ & + \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta) \int_y^{\bar{y}} \left(W(\phi^{\text{poach}}, x, p', \varepsilon', g'_\varepsilon, z') - W(w', y, p', \varepsilon', g'_\varepsilon, z') \right) dF(x) \end{aligned}$$

Optimal search effort will be a function of the cutoff firm $q(\cdot)$; as $q(\cdot)$ falls, the share of offers which result in a higher wage from renegotiation increases, raising the returns to search effort. Note that, while the share of offers which entail a job-to-job transition is fixed at the share of firms with higher productivity (which is independent of the wage), the job-to-job transition rate will be a function of the real wage through search effort decisions.

The privately optimal search effort decision is distinct from the search choice under bilaterally chosen search effort (Faberman et al. (2022)). We model search effort as chosen by the worker, rather than in a bilateral fashion, so that the theory can replicate the fact that search effort is a function of inflation (Pilossoph and Ryngaert (2025), Guerreiro et al. (2025)). If search effort is chosen in a bilateral fashion, the search policy is independent of the real wage. We discuss this technical point in more detail in Section 2.3.

Given the behavior of workers, we can now derive the value of employment to the firm as:

$$\begin{aligned}
J(w, y, p, \varepsilon, g_\varepsilon, z) &= Y(z, y) - \frac{w}{p \cdot \varepsilon} \\
&+ \beta(1 - \delta)(\lambda_e + s_e^*(w, y, p, \varepsilon, g_\varepsilon, z)) \int_{q(w', y, p', \varepsilon', g'_\varepsilon, z')}^y J(\phi^{\text{reneg}}(y, x, p', \varepsilon', g'_\varepsilon, z'), y, p', \varepsilon', g'_\varepsilon, z') dF(x) \\
&+ \beta(1 - \delta)(\lambda_e + s_e^*(w, y, p, \varepsilon, g_\varepsilon, z)) \mathbf{E}_{g'_\varepsilon} \left[\int^{q(w', y, p', \varepsilon', g'_\varepsilon, z')} dF(x) \right] J(w', y, p', \varepsilon', g'_\varepsilon, z') \\
&+ \beta(1 - \delta) \left[(1 - (\lambda_e + s_e^*(w, y, p, \varepsilon, g_\varepsilon, z))) \right] \mathbf{E}_{g'_\varepsilon} J(w', y, p', \varepsilon', g'_\varepsilon, z')
\end{aligned}$$

where LOM1-LOM4 hold and the firm takes the worker's optimal search effort defined in OS1 and OS2 as given.

The firm produces output $Y(z, y) = zy$, but must pay its worker $\frac{w}{p \cdot \varepsilon}$. It then takes as given the worker's search behavior, $s_e^*(w, y, p, \varepsilon, g_\varepsilon, z)$ which determines the probability that the relationship remains intact, and the probability that wages will be renegotiated in the case that it does. In the event of a separation, we assume that the firm earns a value of zero.¹⁰

2.2 Balanced Growth

Proposition 1 *If (i) $Y(z, y)$, $B(z)$, $C(z, s)$ are homogeneous of degree 1 (HOD1) in z and (ii) wage contracts are indexed to trend inflation g_p and productivity growth g_z , then*

$$U((1 + g_z) \cdot z) = (1 + g_z) \cdot U(z),$$

$$W((1 + g_z)(1 + g_p) \cdot w, y, (1 + g_p) \cdot p, \varepsilon, g_\varepsilon, (1 + g_z) \cdot z) = (1 + g_z) \cdot W(w, y, p, \varepsilon, g_\varepsilon, z), \text{ and}$$

$$J((1 + g_z)(1 + g_p) \cdot w, y, (1 + g_p) \cdot p, \varepsilon, g_\varepsilon, (1 + g_z) \cdot z) = (1 + g_z) \cdot J(w, y, p, \varepsilon, g_\varepsilon, z)$$

Proof. See Appendix A ■ Proposition 1 implies that the model scales with TFP growth and trend inflation, so that we can solve the model for a single pair (p, z) , reducing the state space to nominal wages, firm productivity, the stochastic price level component ε , and the shock to the price level g_ε .

Critically, the rate of trend inflation does not matter for allocations as long as it is deterministic and wages are indexed appropriately. The welfare costs of inflation in our model, therefore, arise entirely from unanticipated movements around trend inflation, not from trend inflation itself. Anticipated inflation may therefore impose costs primarily through channels that lie outside our framework; we instead isolate the labor-market channel as one that operates specifically through inflation surprises.

¹⁰While we do not model entry, this is what the value of a vacancy would be in the case of free entry.

In what follows, we will use functions $Y(z, y)$, $B(z)$ and $C(z, s)$ which satisfy the HOD1 property so that we can drop the aggregate price level and aggregate TFP (p, z) from the set of state variables. Specifically, we will assume that:

$$Y(y, z) = z \cdot y, \quad B(z) = z \cdot b, \quad \text{and} \quad C(z, s) = z \cdot c(s)$$

and condense notation by dropping (p, z) from the state-space of the value functions.

2.3 Efficiency

To understand the efficiency properties of the equilibrium, consider the allocation chosen by a planner who can directly control search effort to maximize the joint aggregate value of each match, $M^P(y)$.¹¹

$$\begin{aligned} M^P(y) = \max_{s \in [0, \bar{s}_e]} & y - c(s) + \beta \delta M^P(\emptyset) + \beta(1 - \delta)(s + \lambda_e) \int_{\underline{y}}^y M^P(y) dF(x) \\ & + \beta(1 - \delta)(s + \lambda_e) \int_y^{\bar{y}} M^P(x) dF(x) + \beta(1 - \delta)(1 - s - \lambda_e) M^P(y) \end{aligned} \quad (\text{SP1})$$

where the value of unemployment under the planner's allocation is:

$$M^P(\emptyset) = \max_{s \in [0, \bar{s}_u]} \left\{ b - c(s) + \beta(\lambda_u + s) \int_{\underline{y}}^{\bar{y}} M^P(x) dF(x) + \beta(1 - \lambda_u - s) M^P(\emptyset) \right\} \quad (\text{SP2})$$

The planner's chosen search effort for employed workers is determined by the first-order condition:

$$s_e^{P*}(y) = \begin{cases} 0, & \text{if } \tilde{s}_e^P(y) < 0, \\ \tilde{s}_e^P(y), & \text{if } 0 \leq \tilde{s}_e^P(y) \leq \bar{s}_e, \\ \bar{s}_e, & \text{if } \tilde{s}_e^P(y) > \bar{s}_e \end{cases} \quad (\text{SP3})$$

where

$$\frac{\partial c(s)}{\partial s} \Big|_{\tilde{s}_e^P(y)} = \beta(1 - \delta) \int_{\underline{y}}^y M^P(y) dF(x) + \beta(1 - \delta) \int_y^{\bar{y}} M^P(x) dF(x) - \beta(1 - \delta) M^P(y) \quad (\text{SP4})$$

The planner's search choice differs from the worker's individual choice because it accounts for

¹¹For a complete derivation, see Appendix B. We solve for the planner's joint value first by considering an alternative benchmark of bilaterally efficiency. For more details, see Appendix B.1.

the impact of search on the full match value rather than the workers' value only. The full match value additionally includes the effects of worker search on both (i) the current firm and (ii) future possible values of match output; neither of these are considered when choices are privately optimal from the workers' perspective.¹²

The former consideration would make the planner lean towards lower search effort levels relative to equilibrium choices, since the planner cares about the lost value to the current firm should the worker's search be successful. The latter consideration would make the planner lean towards higher levels of search effort, since the worker does not receive the full value of future matches. Which force dominates depends on the worker's position on the job ladder. At the bottom of the ladder, future match-output gains are large and largely uninternalized by the worker, so privately optimal search is too low relative to the planner's desired level. At the top of the ladder, there is no further productivity gain from moving; there is only the prospect of rent extraction from the current firm, so the worker's incentive to search reflects a transfer the planner does not value, and privately optimal search is too high.

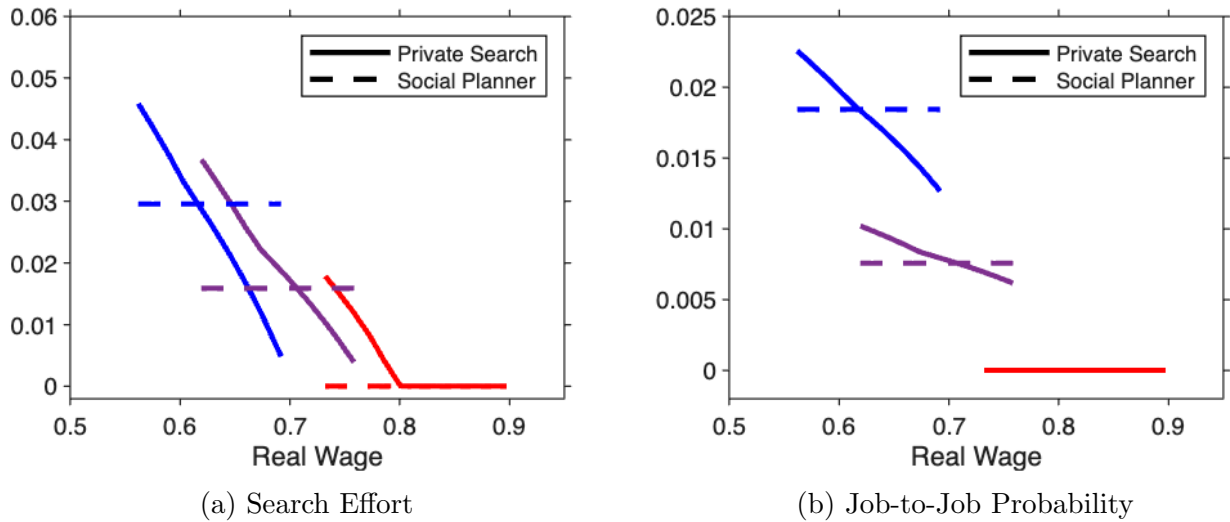


Figure 2: Private vs. Socially Optimal Outcomes

Note: Panel (a) plots privately optimal search effort (solid lines) and the social planner's search effort (dashed lines) against the real wage, for workers employed at firms of different productivity levels; socially optimal search is invariant to wage levels within each firm type. Panel (b) plots the corresponding job-to-job transition probabilities, which under the social planner's policy similarly remain constant across real wage levels for each firm productivity type. Each color corresponds to a different productivity level y (blue, purple, red from lowest to highest, respectively); solid lines show private search effort and dashed lines show the social planner's choice.

¹²Appendix B outlines the optimal level of search under bilaterally efficient search. In this case, the only consideration that a planner has that makes the planner's search deviate from bilaterally efficient search is future match output.

Figure 2 illustrates these forces using the calibrated model. Panel (a) plots privately optimal search effort (solid lines) and the planner’s search effort (dashed lines) against the real wage, separately for workers at firms of different productivity levels. Panel (b) plots the corresponding job-to-job transition probabilities. When search is privately chosen, workers intensify search in response to real-wage erosion, generating a steep negative relationship between the real wage and search effort within a given firm. The planner, by contrast, treats wage levels as transfers within the match. What determines socially optimal search is the trade-off between search costs and the expected productivity gains from reallocation, neither of which depends on how match surplus is divided between worker and firm. The planner’s search policy is therefore invariant to the real wage within a firm type. As a result, the planner’s search policy is monotonic in firm productivity: search is higher at less productive firms and lower at more productive firms, since the expected gain from reallocation is larger when the worker is further from the top of the ladder. The privately optimal policy need not be monotonic in firm productivity. As Panel (a) shows, a worker at a more productive firm whose real wage has been eroded can search more than a worker at a less productive firm whose real wage is intact. Workers at productive firms search at least as much as the planner would regardless of real wage, while workers at less productive firms search too little when their real wages are high and too much when their real wages are low.

3 Calibration

3.1 Parameterization and Targeted Moments

We calibrate the model to match moments from the pre-COVID US economy spanning 2000–2019. One model period is one month. We set the discount rate, β , so that it is consistent with an annual interest rate of 5 percent. We set the deterministic TFP growth rate, g_z , to match an annual growth rate of 0.78 percent, reflecting the average annual growth rate of TFP from 2000 through 2019.¹³ We set trend inflation, g_p , to match an average annual inflation rate of 2 percent. Finally, we set the job offer arrival rate of unemployed workers, λ_u , to be 0.34 so that the model is well-aligned with the offer arrival rate observed in the SCE.

The inflation shock process is AR(1) with parameters ρ and σ_ν . We estimate these parameters using the entire history of the CPI price index in the US from 2000 through 2019. First, we define the cyclical component of annual inflation as the annual inflation rate less the average annual inflation rate over this period of 2 percent. We estimate ρ and σ_ν from the autoregressive model of the cyclical component of inflation. This yields an estimate of ρ of around 0.932 and an estimate of σ_ν of around 0.00036. In practice, we discretize the continuous inflation shock

¹³The TFP series is drawn from Fernald (2014). The series adjusts for variable utilization of capital and labor inputs, providing a more accurate measure of TFP growth.

process ε_p characterized by ρ and σ_v^2 . We use the Rouwenhorst method to approximate the process using 37 grid points.¹⁴

For our quantitative implementation of the model, we extend it in three key dimensions. First, we allow the job destruction rate faced by workers to depend on firm productivity following $\delta(y) = \delta_0 + \delta_1 y$ with $\delta_1 \leq 0$. This embeds a “slippery” job ladder (Jarosch (2023)) into our model, which helps with matching wage growth moments well.¹⁵ We target δ_0 to match the average unemployment rate of 0.068 from the SCE (Faberman et al., 2022), yielding an estimate of δ_0 around 0.0181. We target δ_1 to match the separation-wage elasticity of -0.0392 documented by Jung and Kuhn (2018) using the SIPP, yielding an estimate of -0.022 . Second, we allow there to be exogenous job mobility shocks (“godfather shocks”), λ_g , under which a worker draws a job offer and either can accept the offer or receive the value of unemployment - this essentially alters the worker’s bargaining position with firms for a share λ_g of cases. In the spirit of Faberman et al. (2022), we discipline λ_g using the offer acceptance rate of employed workers, as measured in the SCE¹⁶. The key intuition is that, all else constant, λ_g will tend to increase the share of offers accepted by employed workers. Finally, we introduce moving costs η_e for E-to-E transitions and η_u for UE transitions as described in Appendix D. For the moving costs, we target both the average 12-month real wage growth of job movers and job movers who experience a job separation.¹⁷ The 12-month real wage growth of job movers is measured from administrative payroll data as 0.06 (Grigsby et al., 2021), while the real wage growth of job movers who experience a job separation between the month they lost their job and the month they were re-employed is measured from the SIPP as -0.051 (Gertler et al., 2020). Intuitively, if moving costs are higher, workers are more selective in the job offers they take: conditional on moving, wage growth will be higher the higher are moving costs.

We assume that the search cost function takes the form $C(z, s) = zc(s)$, where $c(s) = c_0 s^\kappa$, $\kappa > 1$. We calibrate κ to match the search effort-wage elasticity of -0.07 documented by Faberman et al. (2022) using the Survey of Consumer Expectations (SCE), where search effort is captured by the decision to submit a job application. Our estimate for κ is 2.27, implying that search costs are approximately quadratic. We calibrate c_0 to match the relative offer yield ratio between employed and unemployed workers of 0.237 documented by Faberman et al. (2022) using the SCE, finding that c_0 is 13.94. Intuitively, the relative offer yield ratio between employed and unemployed workers is closely related with c_0 , as unemployed workers will never put forth search effort in our model and search effort is decreasing in c_0 for employed workers. We use

¹⁴To capture the dynamics of larger inflationary shocks well, we select a large grid.

¹⁵Since separation rates and firm productivity will be perfectly negatively correlated, the ranking of firms (for workers and the planner) remains in order of firm productivity.

¹⁶In our model, unemployed workers accept every offer made to them so we use the acceptance rate of offers by employed workers instead of the acceptance ratio between employed and unemployed workers.

¹⁷The 12-month average real wage growth of job movers includes both workers who experience a non-employment spell before being hired by a new firm, as well as workers who experience a job-to-job transition.

the average monthly job-to-job transition rate in the SCE, found to be 0.025 by Faberman et al. (2022), to discipline the job offer arrival rate, λ_e , faced by employed workers, arriving at $\lambda_e = 0.047$.

We assume that the exogenous vacancy distribution across firm types is a truncated Beta so that $y \sim \text{Beta}(\tilde{\alpha}, \tilde{\beta})$, with support $[b, 1]$. In order to discipline parameters governing the vacancy distribution, we target dispersion in firm fixed effects¹⁸ and the average 12-month real wage growth of job stayers respectively. The empirical moments we target are measured from administrative payroll data as 0.019 for wage growth of job stayers (Grigsby et al., 2021) and 0.032 for dispersion of firm fixed effects (Lachowska et al., 2023). Our estimate for $\tilde{\alpha}$ is 2.98 and our estimate for $\tilde{\beta}$ is 2.68.

Finally, to calibrate the flow value of unemployment b , we target the replacement rate of average flow output, $\frac{b}{\mathbb{E}[y]}$, using a target of 0.55 consistent with exercises in Chodorow-Reich and Karabarbounis (2016). Given the wide range of structural estimates of the replacement rate found in the literature, we place comparatively less weight on this target, allowing for wage growth moments to inform b through its influence on the vacancy distribution. This yields an estimate of b near 0.41.

Table 1 displays our parameter estimates, along with the data source underlying each targeted moment. Table 2 displays model moments together with targeted moments, providing a sense of the fit of the model.

Table 1: Calibrated Parameter Values

Parameter	Description	Value	Source/Target
<i>A. Externally calibrated parameters</i>			
β	discount rate	0.996	5% annual interest rate
g_z	deterministic TFP growth	0.0006	mean TFP growth: 0.78%
λ_u	job arrival rate, unemployed	0.342	SCE
g_p	trend inflation	0.0017	mean CPI inflation (2000-2019)
ρ	persistence of inflation shock	0.932	cyclical CPI inflation (2000-2019)
σ_ν	standard deviation of inflation shock	0.00036	cyclical CPI inflation (2000-2019)
<i>B. Internally calibrated parameters</i>			
κ	elasticity of search cost	2.27	search-wage elasticity
c_0	Search cost parameter	13.94	relative offer yield ratio
λ_e	offer arrival, employed	0.047	EE transition rate
b	flow value of unemployment	0.409	replacement rate
δ_0	job destruction intercept	0.0181	unemployment rate
δ_1	job destruction slope	-0.022	separation-wage elasticity
λ_g	exogenous job mobility shock	0.009	acceptance rate
η_u	moving cost share of non-employed	0.74	mean wage growth of ENE movers
η_e	moving cost share of employed	0.22	mean wage growth of movers
$\tilde{\alpha}, \tilde{\beta}$	vacancy distribution, $\beta(\tilde{\alpha}, \tilde{\beta})$	2.98, 2.68	firm FE variance, stayer wage growth

¹⁸In order to measure dispersion in firm fixed effects in simulated data, we recover fixed effects from a regression of log wages on a firm dummy, and then compute the variance of the estimated firm dummy coefficients.

Our model accurately captures the acceptance rate, where employed workers accept about 32.6% of job offers, closely matching the empirical observation of 32.8%. The unemployment rate in our model is 6.6%, slightly below but still close to the empirical target of 6.8%. Importantly, our model successfully replicates the negative separation-wage elasticity documented in Jung and Kuhn (2018), indicating that higher-wage workers have lower separation rates.

The model performs well in matching wage dynamics across different labor market transitions. This is a key moment for our model, since our theory ties together search effort, inflation, and wage growth. For job stayers, we nearly match the annual real wage growth of about 2%. For job movers, our model nearly matches the empirical 6% annual real wage growth, capturing the substantial wage premium associated with voluntary job transitions. We also closely match the wage penalty experienced by workers who move through unemployment (ENE transitions), with our model generating a 5.6% wage decline compared to the empirical target of 5.1%. Equally as important, the model matches the elasticity of search effort with respect to the real wage. Finally, our model generates a realistic degree of wage dispersion across firms, with the variance of firm fixed effects of 0.034 comparable to its empirical counterpart of 0.032, drawn from Lachowska et al. (2023).

Table 2: Targeted Moments in the Data and the Model

Moment	Source	Data	Model
search-wage elasticity	Faberman et al. (2022)	−0.07	−0.07
offer yield	Faberman et al. (2022)	0.237	0.232
acceptance rate	Faberman et al. (2022)	0.328	0.326
EE transition rate	Faberman et al. (2022)	0.025	0.027
replacement rate	Chodorow-Reich and Karabarbounis (2016)	0.55	0.55
Unemployment rate	Faberman et al. (2022)	0.068	0.066
separation-wage elasticity	Jung and Kuhn (2018)	−0.0392	−0.0393
real wage growth (job stayers)	Grigsby et al. (2021)	0.019	0.023
real wage growth (job movers)	Grigsby et al. (2021)	0.06	0.057
real wage growth (ENE movers)	Gertler et al. (2020)	−0.051	−0.056
dispersion in Firm FEs	Lachowska et al. (2023)	0.032	0.034

4 The Search Costs of Inflation in the Labor Market

In this section, we use our estimated model to quantify the costs to workers of unexpected shocks to inflation in two ways. First, we consider a realization of g_ε whose magnitude generates a movement in overall inflation similar to what was experienced in the post-COVID period. Second, we compare the baseline economy to an economy without shocks ($\sigma_\nu^2 = 0$) which is the same as comparing the estimated model to an alternative environment where wages are fully indexed to the price level.

4.1 The Costs of the COVID Inflation Episode

We consider a one-time, unanticipated, temporary movement in inflation precipitated by a shock ν at some date τ that mimics the inflation dynamics of the COVID period. Specifically, ε moves from its mean to a value that generates an annualized inflation rate of $p\varepsilon^{shock}$ at around 9 percent at period τ .¹⁹ From $\tau + 1$ onward, ε evolves according to the estimated AR(1) process; agents observe the shock when it occurs and form expectations about its future path using the same process they used before τ . That is, the shock itself is unanticipated, but its persistence is not.²⁰

Figures 3a and 3b plot the price level (normalized at time $\tau - 1$) and the inflation rate. Although the rise in inflation is transitory (annualized inflation falls from roughly 9 to 4 percent within two years and returns to trend after about six) it produces a permanent shift in the price level. This permanence is what matters for nominal wage contracts: real wages do not mechanically revert when inflation returns to trend.

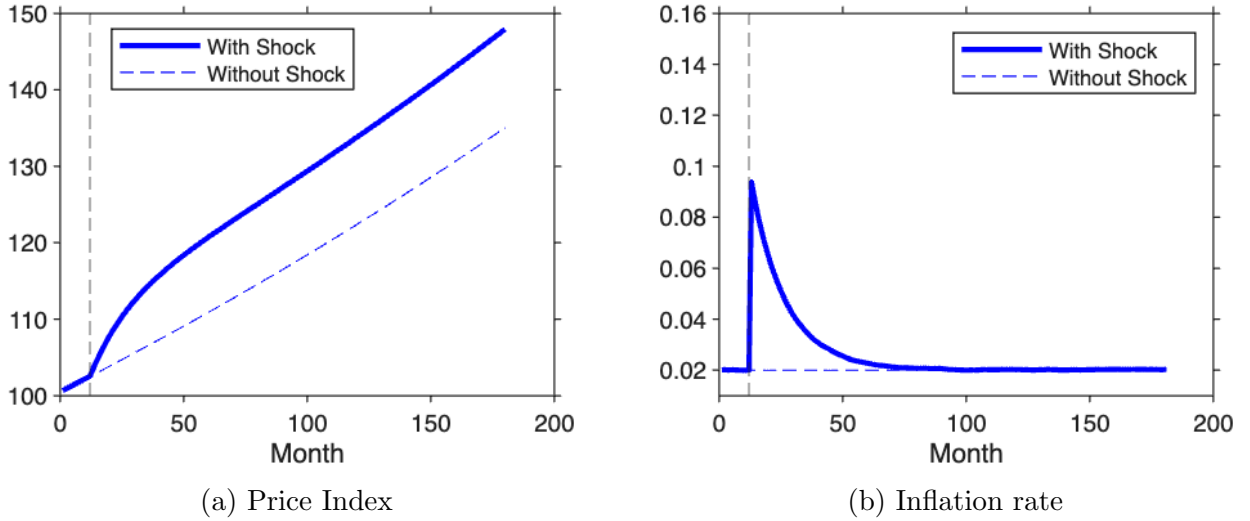


Figure 3: Graphical Illustration of the Inflation Shock

Note: Figure 3a plots the realized price index at time t , normalized relative to the first period of the plot, in both the environment absent the shock (dashed line) and the counterfactual which introduces the shock (solid line). Figure 3b plots the corresponding implied inflation rate in the two scenarios, highlighting the magnitude of the shock.

Because contracted wages are not indexed to surprise inflation, real wages drop on impact.

¹⁹While agents believe that a shock of this magnitude is possible, it is extremely unlikely. In this sense, it is appropriate to describe the shock as being unexpected.

²⁰In practice, we simulate a large set of economies under different inflation paths. To do this, we draw a set of inflation paths where ε is equal to its mean in $\tau - 1$ and then equal to ε^{shock} at τ , drawing future inflation paths for T periods given the transition matrix governing ε . We simulate all objects of interest implied by our model under each inflation path, taking the average over inflation paths to consider the effect of the shock.

In the two years following the shock, average real wages decline by roughly 2 percentage points relative to the no-shock path (Figure 4a). Workers respond by raising search effort by 30 percent on impact (Figure 4b), which slowly dissipates as nominal wages are bid up through outside offers.

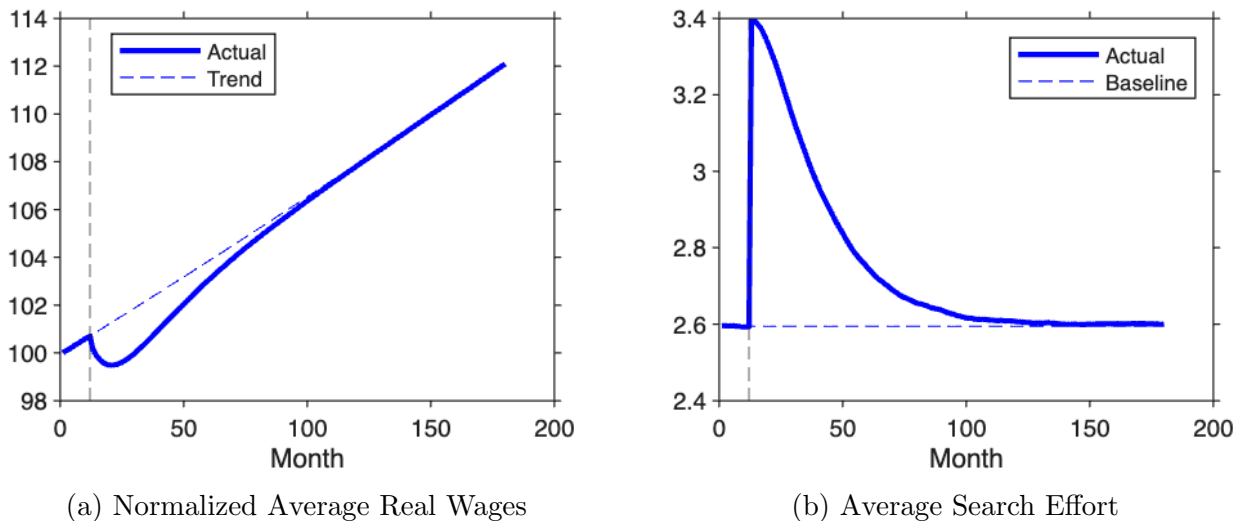


Figure 4: Average Real Wages and Search Effort Following a Transitory Shock to Inflation

Note: Figure 4a plots the index of average real wages at time t , normalized relative to $\tau - 1$, in response to the inflationary shock, along with the same object in the absence of the shock. Figure 4b plots average search effort over time in response to the inflationary shock, along with the same object in the absence of the shock.

This heightened search generates wage growth through two channels (Figures 5a and 5b). First, additional search contacts translate into higher job-to-job transition rates. Second, a larger share of offers triggers renegotiation, so workers who remain with their employer also obtain wage increases.

Figure 6 reports the average welfare loss from the shock in consumption-equivalent terms.²¹ Workers would give up 0.44 percent of flow consumption to avoid the shock. Of this loss, roughly 73 percent reflects lower real wages, 14 percent the costs of additional search, and 13 percent the costs of additional mobility. The magnitude of these costs sits between two familiar benchmarks: it is roughly an order of magnitude larger than Lucas’s (2003) estimate of the welfare cost of US business cycle fluctuations, but only about one-tenth the welfare cost of an individual experiencing job displacement in our model. The latter comparison reflects the fact that this is the cost borne by the average worker — most workers absorb modest losses, while

²¹Since utility is linear, this is equivalent to the percent decline in the present discounted value of income net of search and moving costs.

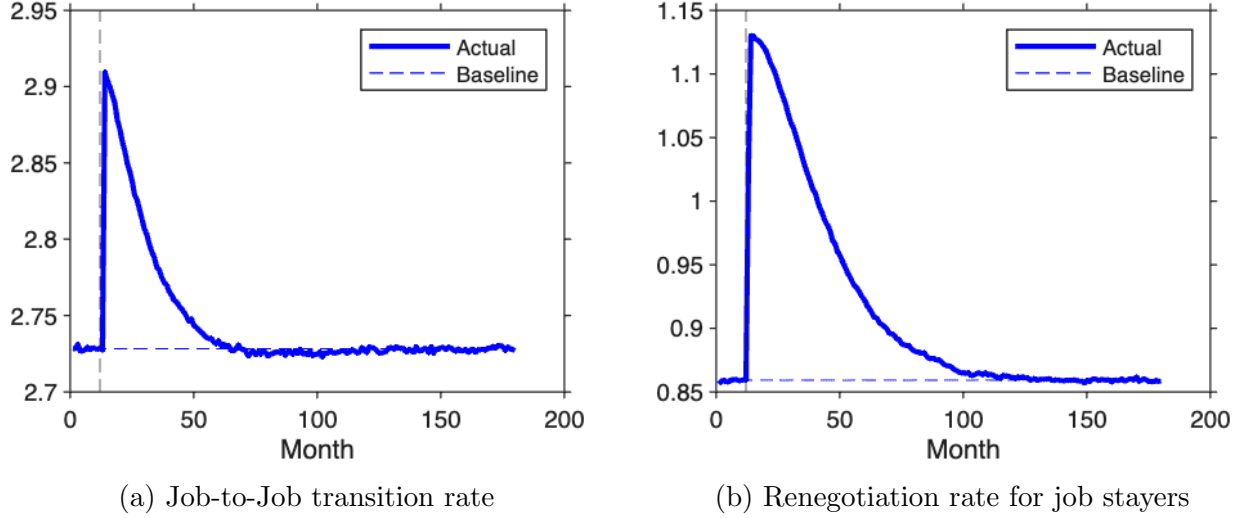


Figure 5: Mechanisms of Adjustment

Note: Figure 5a plots the job-to-job transition rate in response to the inflationary shock (blue solid line), along with a dashed line indicating the steady-state job-to-job transition rate. Figure 5b plots the rate at which employed workers who are job stayers renegotiate their contracts using outside offers in response to the inflationary shock (blue solid line), along with a dashed line indicating the steady-state renegotiation rate.

job loss is a concentrated cost borne by the displaced.

This average masks substantial heterogeneity. The discussion in Section 2 makes clear that search propensity depends on both nominal and real wages, so an inflation shock is experienced differently across the wage distribution. Sorting workers into quartiles based on income in period $\tau - 1$ and following each group forward, Figure 7 shows that workers at the top of the distribution bear the largest losses - over six times those at the bottom - and nearly all of their loss reflects foregone real wages. Workers at the bottom, by contrast, have the most to gain from search and therefore pay the search and moving costs to improve their position; high-wage workers effectively absorb the real wage losses instead.

4.1.1 From Worker Losses to Net Social Costs

Worker welfare losses overstate the aggregate cost of the shock for two reasons. First, lower real wages transfer surplus from workers to firms. Second, the increased search effort reallocates workers to more productive matches, raising aggregate output. Both channels partially offset the worker-side costs.

Figure 8 reports the average present discounted value of aggregate welfare losses from the shock. The per-period losses from higher search and moving costs aggregate to a nearly 0.10 percent reduction in welfare relative to baseline. This loss, however, is almost entirely offset by

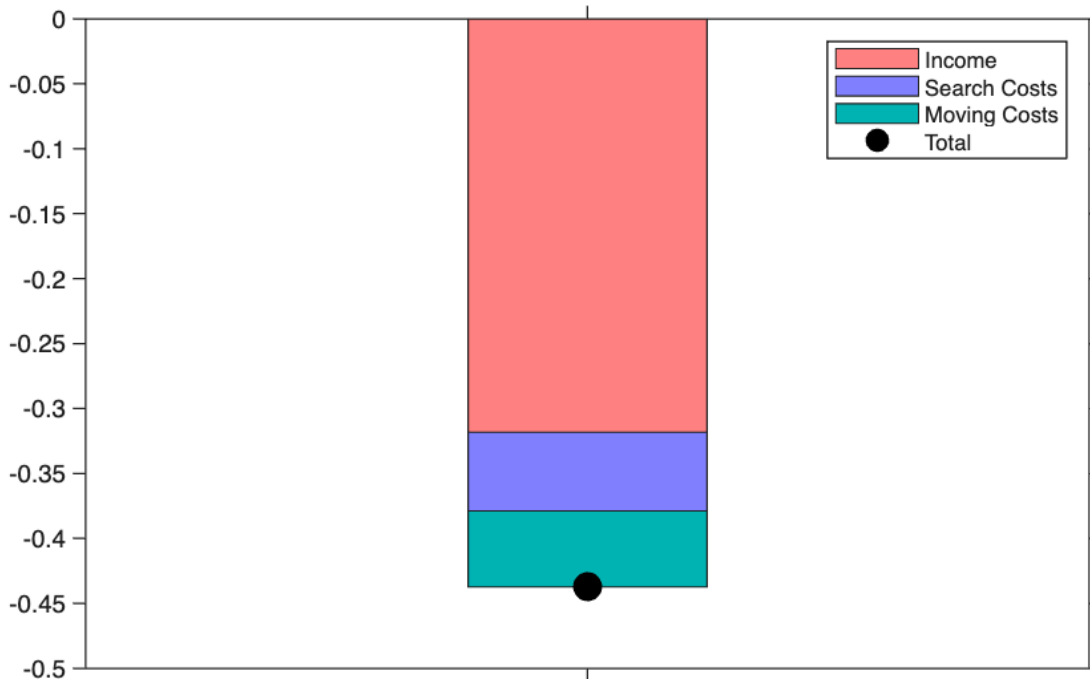


Figure 6: Lifetime Worker Welfare Costs of an Inflationary Shock

Note: Figure 6 plots the welfare losses (measured as consumption equivalents) as a black dot. The colored bars decompose this difference into a component explained by the difference in income, a component explained by the difference in incurred search costs, and a component explained by the difference in incurred moving costs.

the PDV of the per-period output gains, leaving a net welfare cost of just 0.004 percent.

One concern with this calculation is that we have held the passive offer arrival rate λ_e fixed, treating the increase in job-to-job transitions as operating entirely through endogenous worker search effort. In practice, vacancy posting and contact rates may co-move with inflation, amplifying the reallocation channel. Appendix F extends the model to allow λ_e to move deterministically with the cyclical component of inflation, calibrated to match the empirical co-movement between the job-finding rate and detrended inflation. Under this extension, the net social effect of the COVID-sized shock flips from a small loss to a small gain of roughly 0.13 percent, driven by additional output from accelerated reallocation up the job ladder. Worker welfare losses also fall, since the higher arrival rate substitutes for some of the search effort workers would otherwise exert. The qualitative finding that worker losses substantially overstate aggregate losses, however, is robust.

Taken together, these results suggest that a single large inflationary episode imposes meaningful costs on workers but only modest net costs on the economy as a whole. The aggregate

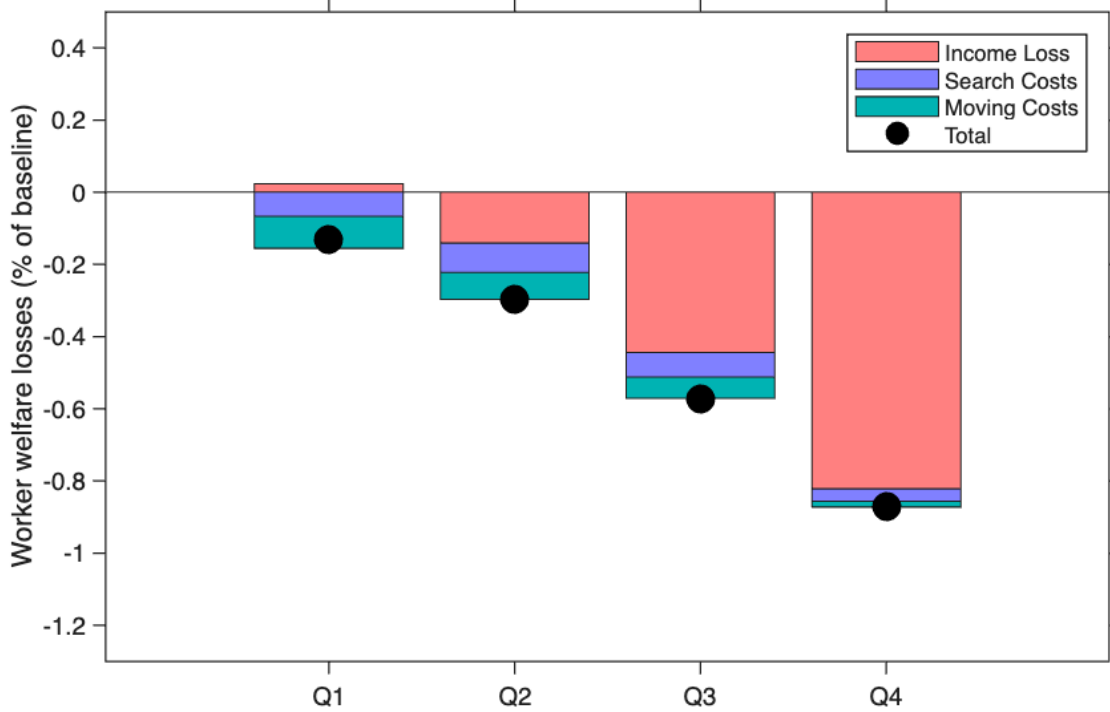


Figure 7: Lifetime Welfare Consequences by pre-shock Wage Quartile

Note: Figure 7 plots the welfare losses (measured as consumption equivalents) as a black dot, separately for workers in different wage quartiles as of $\tau - 1$ (pre-shock). The colored bars decompose this difference into a component explained by the difference in income, a component explained by the difference in incurred search costs, and a component explained by the difference in incurred moving costs.

picture changes, however, when we consider ongoing inflation volatility rather than a one-time shock. As we show next, the wedge between worker costs and social costs that produces the small net cost here generates a meaningfully different result when search effort is sustained at elevated levels over the long run.

4.2 The Costs of Cyclical Shocks

We now turn from the welfare consequences of a single large inflation shock to those of ongoing inflation volatility around the trend rate. We can loosely think of the inflation process as the outcome of monetary policy, in which the central bank aims to keep inflation close to its 2% target but fails to do so perfectly. The magnitude of these deviations determines σ_ν , and in this section we ask how the welfare properties of the labor market vary with σ_ν .

Throughout, we compare our estimated model to a baseline scenario of perfect price stability ($\sigma_\nu^2 = 0$), which is isomorphic to an environment where inflation fluctuates but wages are fully

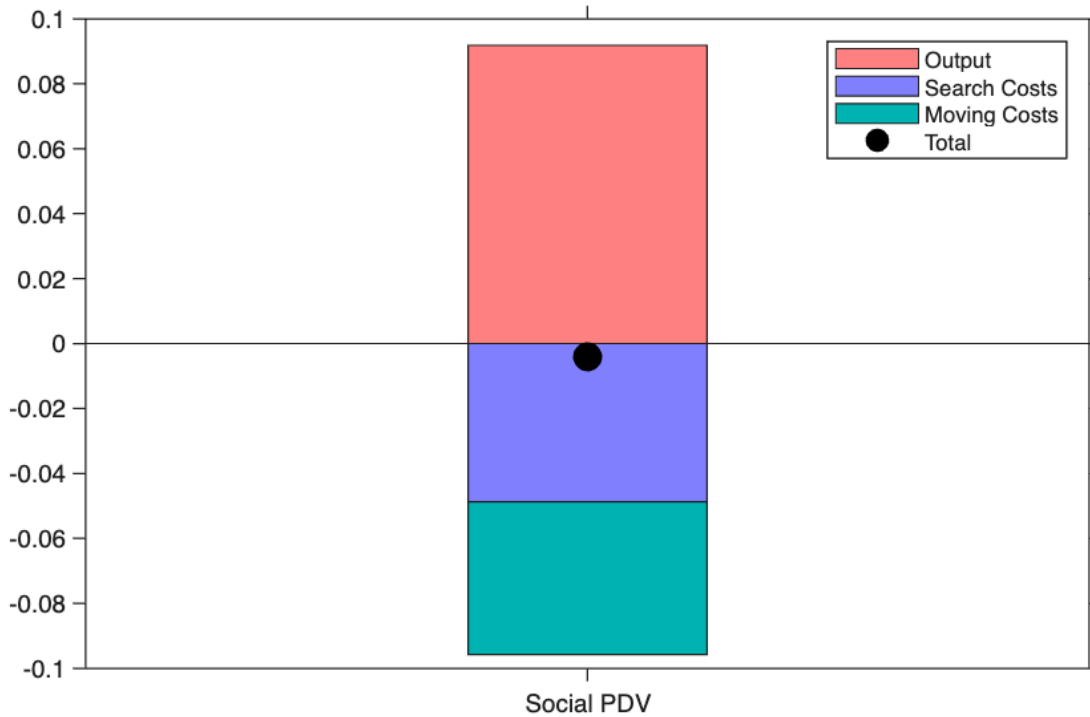


Figure 8: Welfare Costs of an Inflationary Shock

Note: Figure 8 plots the mean percent difference in the present discounted value of output net of search and moving costs in the shocked economy and the baseline one as a black dot. The colored bars decompose this difference in PDVs into a component explained by the difference in PDV of output, a component explained by the difference in the PDV of search costs, and a component explained by the difference in the PDV of moving costs.

indexed to the realized price level rather than just the trend. We vary σ_ν from near-zero to levels substantially exceeding those observed in the US, holding all other parameters fixed at their calibrated values.

4.2.1 Worker and Social Welfare

Figure 9 reports worker and social welfare as a function of inflation volatility, both expressed relative to the perfect price stability baseline. Panel (a) shows that worker welfare is approximately flat across the range of volatility we consider. Panel (b) shows that social welfare, by contrast, declines monotonically and meaningfully with volatility.

The gap between these two panels is informative. Whenever contracts are renegotiated, firms offer initial wages that compensate workers for the expected search costs they will incur over the course of the match, so that, for example, a worker hired out of unemployment is indifferent between employment and unemployment regardless of the volatility regime; worker

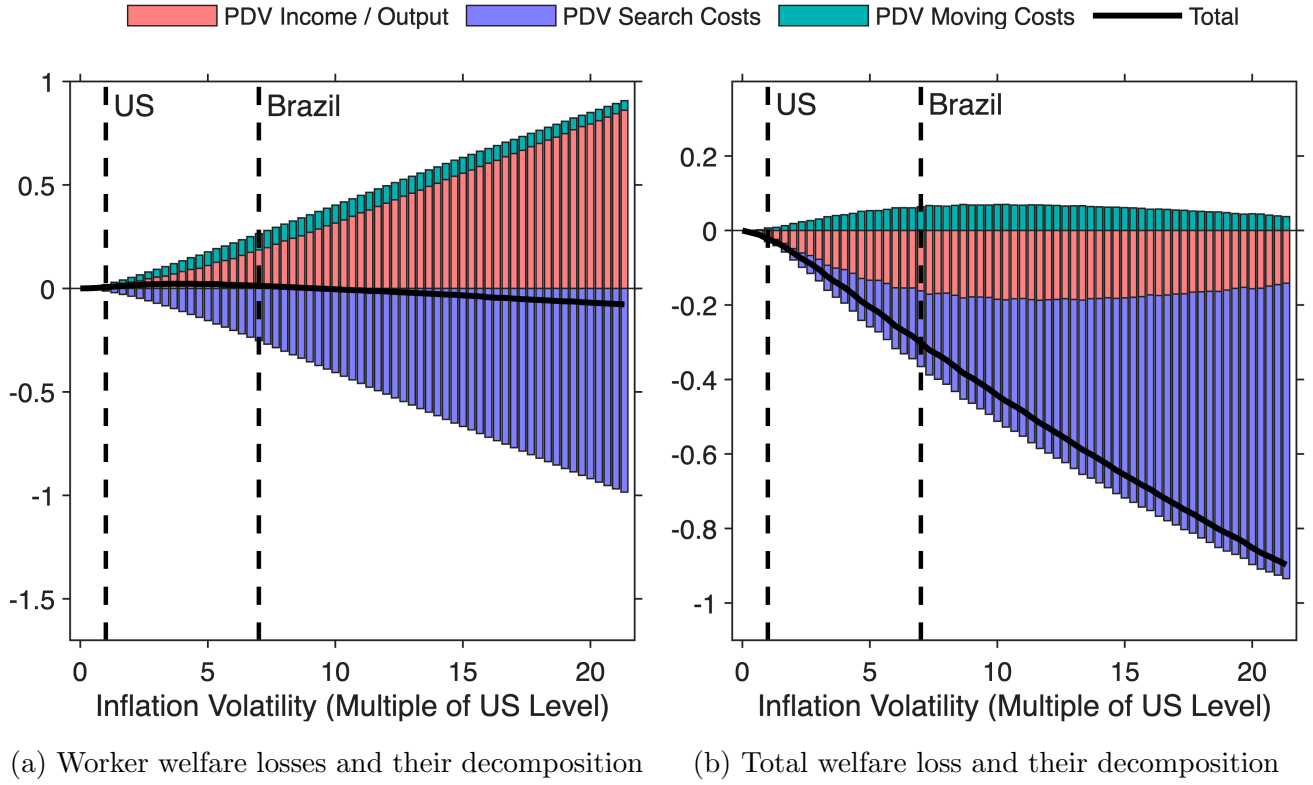


Figure 9: Welfare Losses from Inflation Volatility

Note: Panel (a) decomposes the *worker* welfare difference between each volatility level and the perfect price stability baseline. The teal bar captures the contribution of differences in moving costs. The red bar captures the contribution of differences in income; the blue bar captures the contribution of differences in search costs. The black line shows the total welfare loss. Panel (b) shows the social welfare loss (as a percentage of baseline welfare) from inflation volatility relative to perfect price stability. The red bars represent output losses here rather than income differences, since those are just transfers between workers and employers from the planner's perspective. In both panels, the x-axis represents inflation volatility as a multiple of the US level, all values are expressed as percentages of baseline welfare, and the vertical dashed lines mark *approximate* country comparisons.

welfare losses from higher volatility are therefore small - higher average search costs are offset by higher average wages. Social welfare, however, internalizes the deadweight loss from search effort itself. As volatility rises, aggregate search costs rise even when workers are approximately compensated for them, and these costs show up as reduced profits for firms and reduced output net of search.

Figure 9 decomposes the social welfare loss into its components. Search costs rise sharply with volatility and account for essentially all of the decline; output and moving costs contribute only modestly. The workers who bear the welfare losses in any individual realization are compensated in equilibrium through higher wages, but the search effort they exert is a real social cost that firms pay in lower profits and that no amount of compensation can offset.

Inflation volatility is not costless simply because workers appear roughly indifferent to it. Its costs are borne by firms in the first instance, and by society in aggregate, through the search activity that workers undertake to protect themselves against real wage declines. At US levels of volatility, the social welfare cost is small — around 0.03% of baseline. At Brazil-level volatility, several times higher than the US, the cost rises to roughly 0.3% of baseline, large enough to be a first-order consideration in any cross-country comparison of monetary policy regimes. The stabilization role of monetary policy thus has a labor-market rationale that operates independently of the classic price-dispersion arguments (e.g., Nakamura et al. (2018)): reducing inflation volatility reduces the volatility of real wages, which in turn reduces wasteful search effort.

5 Conclusion

This paper develops a theory of the labor market costs of inflation. When wages are contracted in nominal terms and adjust only through costly search and renegotiation, unanticipated inflation imposes a trade-off on workers: accept the loss in real income, or pay the costs of search and mobility to avoid it. We extend the Postel-Vinay and Robin (2002) model to incorporate nominal contracting and private search effort, and use the estimated model to quantify the resulting welfare costs along two dimensions.

Our first exercise considers a one-time, COVID-sized inflation shock. We find an average welfare loss of 0.44% of flow consumption, with substantial heterogeneity: workers at the top of the wage distribution bear losses over six times those at the bottom. Once the offsetting gains to firms and from productive reallocation are netted out, the social cost of the shock is two orders of magnitude smaller. Our second exercise examines inflation volatility. Worker welfare is approximately flat in inflation volatility, because firms compensate workers at hiring for the expected search costs they will bear. On the other hand, social welfare declines with inflation volatility because those search costs are a resource cost borne by firms and by aggregate output.

Two extensions are worth pursuing. First, worker utility is linear in consumption, so the welfare numbers abstract from risk aversion; they reflect the resource costs of search and the redistribution of real wages between workers and firms, but not any premium for bearing consumption volatility. Adding curvature would raise these costs, especially in the volatility exercise. Second, we treat the inflation process as exogenous. A fuller account would endogenize the central bank’s response and firms’ price-setting; our results suggest the labor-market channel belongs in such an analysis, since it provides a stabilization rationale for monetary policy distinct from the standard price-dispersion arguments.

The labor market is a quantitatively important channel for the welfare costs of inflation, complementary to the asset-price effects in the recent literature (Doepke and Schneider (2006b), Del Canto

et al. (2025)). Survey evidence on workers' dislike of inflation is consistent with this: real wage losses and the search response to them impose real costs.

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A Proof of Proposition 1

Proof. We will take a guess and verify approach. Start with the value of unemployment $U(z)$ evaluated at $U((1+g_z)z)$:

$$\begin{aligned} U((1+g_z)z) &= B((1+g_z)z) + \beta U((1+g_z)^2 z) \\ &= (1+g_z)B(z) + (1+g_z)U((1+g_z)z) \\ &= (1+g_z)U(z) \end{aligned}$$

Now let's look at the value of employment to the worker. First, using the implicit function for the renegotiated wage, multiplying both sides by $(1+g_z)$ we know that:

$$(1+g_z)W(\phi^{\text{reneg}}(y_1, y_2, p, \varepsilon, g_\varepsilon, z), y_1, p, \varepsilon, g_\varepsilon, z) = (1+g_z)W(zy_2p, y_2, p, \varepsilon, g_\varepsilon, z)$$

Using our guess for the scaling, the LHS is equivalent to:

$$W((1+g_z)(1+g_p)\phi^{\text{reneg}}(y_1, y_2, p, \varepsilon, g_\varepsilon, z), y_1, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z))$$

Defining

$$\phi^{\text{reneg}}(y_1, y_2, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) = (1+g_z)(1+g_p)\phi^{\text{reneg}}(y_1, y_2, p, \varepsilon, g_\varepsilon, z) \quad (4)$$

implies:

$$W(\phi^{\text{reneg}}(y_1, y_2, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)), y_1, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) = (1+g_z)W(\phi^{\text{reneg}}(y_1, y_2, p, \varepsilon, g_\varepsilon, z), y_1, p, \varepsilon, g_\varepsilon, z)$$

Using similar arguments,

$$W(\phi^{\text{poach}}(y_2, y_1, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)), y_2, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) = (1+g_z)W(\phi^{\text{poach}}(y_2, y_1, p, \varepsilon, g_\varepsilon, z), y_2, p, \varepsilon, g_\varepsilon, z)$$

Finally, using the function which implicitly defines q :

$$W(w(1+g_z)(1+g_p), y_1, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) = (1+g_z)W(w, y_1, p, \varepsilon, g_\varepsilon, z) = (1+g_z)W(zqp, q, p, \varepsilon, g_\varepsilon, z)$$

So q is unchanged. Turning to the value function and letting $\xi = (1-\delta)(s+\lambda_e)$, $\xi' = (1-\delta)(1-s-\lambda_e)$, we have:

$$\begin{aligned} W(w(1+g_z)(1+g_p), y, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) &= \\ \max_{s \in [0, \bar{s}_e]} (1+g_z) \frac{w}{p\varepsilon} - (1+g_z)C(z, s) + \beta(1+g_z)\delta U(z) & \\ + \beta \mathbf{E}_{g'_\varepsilon} \xi \int^{q(w', y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, (1+g_z)^2 z)}^{q(w', y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, (1+g_z)^2 z)} W(w', y, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z) dF(x) & \\ + \beta \mathbf{E}_{g'_\varepsilon} \xi \int_{q(w', y, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z)}^y W(\phi^{\text{reneg}}(y, x, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z), y, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z) dF(x) & \\ + \beta \mathbf{E}_{g'_\varepsilon} \xi \int_y^{\bar{y}} W(\phi^{\text{poach}}(x, y, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z), x, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z) dF(x) & \\ + \beta \mathbf{E}_{g'_\varepsilon} \xi' W(w', y, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z) & \end{aligned}$$

where

$$w' = \begin{cases} \hat{w} : J(\hat{w}, y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) = 0, & \text{if } J(w(1+g_z)^2(1+g_p)^2, y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) < 0 \\ \hat{w} : W(\hat{w}, y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) - U(z(1+g_z)^2) = 0, \\ \text{if } W(w(1+g_z)(1+g_p), y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) - U(z(1+g_z)^2) < 0, \\ w(1+g_z)^2(1+g_p)^2, & \text{else} \end{cases}$$

For the case where $w' = w(1+g_z)^2(1+g_p)^2$,

$$W(w', y, (1+g_p)^2 p, \varepsilon', g'_\varepsilon, (1+g_z)^2 z) = (1+g_z)W(w(1+g_z)(1+g_p), y, (1+g_p)p, \varepsilon', g'_\varepsilon, (1+g_z)z)$$

For the case where $w' = \hat{w}$ is defined implicitly using

$$J(\hat{w}, y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) = 0$$

the scaling conjecture implies:

$$(1+g_z)J(\hat{w}(y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)), y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) = \\ J((1+g_z)(1+g_p)\hat{w}(y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)), y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) = 0$$

so that

$$\hat{w}(y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) = (1+g_z)(1+g_p)\hat{w}(y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z))$$

Similar arguments hold for the case where $w' = \hat{w}$ is defined implicitly using

$$W(\hat{w}, y, p(1+g_p)^2, \varepsilon', g'_\varepsilon, z(1+g_z)^2) - U(z(1+g_z)^2) = 0$$

Applying these results to the value function:

$$\begin{aligned} W(w(1+g_z)(1+g_p), y, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) = \\ \max_{s \in [0, \bar{s}_\varepsilon]} (1+g_z) \frac{w}{p\varepsilon} - (1+g_z)C(z, s) + \beta(1+g_z)\delta U(z) \\ + \beta(1+g_z)\mathbf{E}_{g'_\varepsilon} \xi \int_{q(w', y, p(1+g_p), \varepsilon', g'_\varepsilon, z)}^{q(w', y, p(1+g_p), \varepsilon', g'_\varepsilon, z)} W(w', y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) dF(x) \\ + \beta(1+g_z)\mathbf{E}_{g'_\varepsilon} \xi \int_{q(w', y, p(1+g_p), \varepsilon', g'_\varepsilon, z)}^y W(\phi^{\text{reneg}}(y, x, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)), y, p(1+g_p), \varepsilon', g'_\varepsilon, z) dF(x) \\ + \beta(1+g_z)\mathbf{E}_{g'_\varepsilon} \xi \int_y^{\bar{y}} W(\phi^{\text{poach}}(x, y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)), x, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) dF(x) \\ + \beta(1+g_z)\mathbf{E}_{g'_\varepsilon} \xi' W(w', y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) \end{aligned}$$

where now

$$w' = \begin{cases} \hat{w} : J(\hat{w}, y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) = 0, & \text{if } J(w(1+g_z)(1+g_p), y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) < 0 \\ \hat{w} : W(\hat{w}, y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) - U(z(1+g_z)) = 0, \\ \text{if } W(w(1+g_z)(1+g_p), y, p(1+g_p), \varepsilon', g'_\varepsilon, z(1+g_z)) - U(z(1+g_z)) < 0, \\ w(1+g_z)(1+g_p), & \text{else} \end{cases}$$

Therefore

$$W(w(1+g_z)(1+g_p), y, p(1+g_p), \varepsilon, g_\varepsilon, z(1+g_z)) = (1+g_z)W(w, y, p, \varepsilon, g_\varepsilon, z) \quad (5)$$

so that the p, z arguments can be dropped. Similar arguments hold for the value of a job to the firm. ■

B Planner's Allocation

To solve for the planner's allocation, it is instructive to first consider the bilaterally efficient allocation.

B.1 Bilateral Efficiency

First, we derive the bilaterally efficient degree of search effort, that is when search effort is determined through a contract that maximizes the joint surplus of the worker-firm pair. We again assume the conditions required for balanced growth and drop (p, z) from the state space. Recall the worker value, here for some search effort level s :

$$\begin{aligned} W(w, y, \varepsilon, g_\varepsilon) = & \underbrace{\frac{w}{\varepsilon} - c(s) + \beta\delta U}_{A_0^W} \\ & + \underbrace{\beta \mathbf{E}_{g'_\varepsilon} (1 - \delta)(s + \lambda_e) \times \int_{\underline{y}}^{q(w', y, \varepsilon', g'_\varepsilon)} W(w', y, \varepsilon', g'_\varepsilon) dF(x)}_{A_1^W} \\ & + \underbrace{\beta \mathbf{E}_{g'_\varepsilon} (1 - \delta)(s + \lambda_e) \times \int_{q(w', y, \varepsilon', g'_\varepsilon)}^y W(\phi^{reneg}, y, \varepsilon', g'_\varepsilon) dF(x)}_{A_2^W} \\ & + \underbrace{\beta \mathbf{E}_{g'_\varepsilon} (1 - \delta)(s + \lambda_e) \times \int_y^{\bar{y}} W(\phi^{poach}, x, \varepsilon', g'_\varepsilon) dF(x)}_{A_3^W} \\ & + \underbrace{\beta \mathbf{E}_{g'_\varepsilon} (1 - \delta)(1 - s - \lambda_e) W(w', y, \varepsilon', g'_\varepsilon)}_{A_4^W} \end{aligned} \quad (6)$$

and the firm value, again for some search level s :

$$\begin{aligned}
J(w, y, \varepsilon, g_\varepsilon) = & \underbrace{y - \frac{w}{\varepsilon}}_{A_0^J} \\
& + \underbrace{\beta(1 - \delta)(s + \lambda_e) \mathbf{E}_{g'_\varepsilon} \int_{q(w', y, \varepsilon', g'_\varepsilon)}^y J(\phi^{reneg}, y, \varepsilon', g'_\varepsilon) dF(x)}_{A_2^J} \\
& + \underbrace{\beta(1 - \delta)(s + \lambda_e) \mathbf{E}_{g'_\varepsilon} \left[\int_{\frac{y}{y}}^{q(w', y, \varepsilon', g'_\varepsilon)} dF(x) \right] J(w', y, \varepsilon', g'_\varepsilon)}_{A_1^J} \\
& + \underbrace{\beta(1 - \delta)(1 - s - \lambda_e) \mathbf{E}_{g'_\varepsilon} J(w', y, \varepsilon', g'_\varepsilon)}_{A_4^J}
\end{aligned} \tag{7}$$

Summing together the values, defining $J(w, y, \varepsilon, g_\varepsilon) + W(w, y, \varepsilon, g_\varepsilon) = M(y)$, using the fact that in case 3 of the text:

$$W(\phi^{poach}, x, \varepsilon', g'_\varepsilon) = M(y),$$

and noticing that the price arguments drop out since the real wage term drops out, the joint match value with bilaterally efficient search is then:

$$\begin{aligned}
M^B(y) = & \max_{s \in [0, \bar{s}_e]} \underbrace{y - c(s) + \beta\delta U}_{A_0^W + A_0^J} + \underbrace{\beta(1 - \delta)(s + \lambda_e) \int_{\frac{y}{y}}^y M^B(y) dF(x)}_{A_1^J + A_2^J + A_1^W + A_2^W} \\
& + \beta(1 - \delta)(s + \lambda_e) \times \int_y^{\bar{y}} M^B(y) dF(x) \\
& + \underbrace{\beta(1 - \delta)(1 - s - \lambda_e) M^B(y)}_{A_4^J + A_4^W} \\
= & \max_{s \in [0, \bar{s}_e]} y - c(s) + \beta\delta U + \beta(1 - \delta) M^B(y)
\end{aligned} \tag{8}$$

The bilaterally efficient amount of search for employed workers is therefore zero.

B.2 Planner's Problem

When contracts are such that the worker captures the full value of the match when hired, whether from non-employment or when poached, the bilaterally efficient search choice is *socially* efficient, as search choice internalizes gains from moving up the productivity ladder.

The joint value of match from the planner's perspective is:

$$\begin{aligned}
M^P(y) = & \max_{s \in [0, \bar{s}_e]} y - c(s) + \beta \delta M^P(\emptyset) \\
& + \beta(1 - \delta)(1 - (\lambda_e + s))M^P(y) \\
& + \beta(1 - \delta)(\lambda_e + s) \int_{\underline{y}}^y M^P(y) dF(x) \\
& + \beta(1 - \delta)(\lambda_e + s) \int_y^{\bar{y}} M^P(x) dF(x)
\end{aligned}$$

where value of unemployment is given by:

$$M^P(\emptyset) = \max_{s_u \in [0, \bar{s}_u]} b - c(s) + \beta(\lambda_u + s) \int M^P(x) dF(x) + \beta(1 - \lambda_u - s)M^P(\emptyset) \quad (9)$$

The difference between the planner's joint value and the bilaterally efficient value is that when workers move to new firms, they receive the joint value of the new match rather than the old match, which mechanically internalizes the value of search to the current match. The associated policy functions for search effort can be found in the text, Equations SP3-SP4.

C Social Value of a Match

In exercises where we consider aggregate welfare in the economy where search effort is chosen privately, we consider the value of matches in the economy from the planner's perspective — we call this the *social* value of a match. From the planner's point of view, the value of the match takes the form of SP1, but with search effort choices taken as given, set to maximize the worker's value, as in OS2:

$$\begin{aligned}
\tilde{M}(w, y, \varepsilon, g_\varepsilon) = & y - c(s_e^*(w, y, \varepsilon, g_\varepsilon)) + \beta \delta \mathbf{E}_{g'_\varepsilon} \tilde{U}(\varepsilon', g'_\varepsilon) \\
& + \beta(1 - \delta)(1 - \lambda_g)(s_e^*(w, y, \varepsilon, g_\varepsilon) + \lambda_e) \mathbf{E}_{g'_\varepsilon} \int_{\underline{y}}^{q(w', y, \varepsilon', g'_\varepsilon)} \tilde{M}(w', y, \varepsilon', g'_\varepsilon) dF(x) \\
& + \beta(1 - \delta)(1 - \lambda_g)(s_e^*(w, y, \varepsilon, g_\varepsilon) + \lambda_e) \mathbf{E}_{g'_\varepsilon} \int_{q(w', y, \varepsilon', g'_\varepsilon)}^y \tilde{M}(\phi^{\text{reneg}}, y, \varepsilon', g'_\varepsilon) dF(x) \\
& + \beta(1 - \delta)(1 - \lambda_g)(s_e^*(w, y, \varepsilon, g_\varepsilon) + \lambda_e) \mathbf{E}_{g'_\varepsilon} \int_y^{\bar{y}} \left(\tilde{M}(\phi^{\text{poach}}, x, \varepsilon', g'_\varepsilon) - \eta_e(x) \right) dF(x) \\
& + \beta(1 - \delta)(1 - \lambda_g)(1 - s_e^*(w, y, \varepsilon, g_\varepsilon) - \lambda_e) \mathbf{E}_{g'_\varepsilon} \tilde{M}(w', y, \varepsilon', g'_\varepsilon) \\
& + \beta(1 - \delta) \lambda_g \mathbf{E}_{g'_\varepsilon} \int_{\underline{y}}^{\bar{y}} \left[\tilde{M}(\phi_u(x, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_u(x) \right] dF(x) \quad (10)
\end{aligned}$$

Similarly, the social value of unemployment is the present discounted value of home production

and of the matches formed out of unemployment, net of moving costs:

$$\tilde{U}(\varepsilon, g_\varepsilon) = b + \beta \lambda_u \mathbf{E}_{g'_\varepsilon} \int_{\underline{y}}^{\bar{y}} \left[\tilde{M}(\phi_u(x, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_u(x) \right] dF(x) + \beta(1 - \lambda_u) \mathbf{E}_{g'_\varepsilon} \tilde{U}(\varepsilon', g'_\varepsilon) \quad (11)$$

where we use the result that unemployed workers exert no search effort ($s_u^* = 0$).

Accordingly, the social value of a match depends on the negotiated wage, the inflationary state and the price level, as all three of these factors are inputs into the worker's choice of search effort.

D Extended Quantitative Model

For our quantitative exercise, we extend the model in three key dimensions. First, we allow the job destruction rate faced by workers to depend on firm productivity: $\delta(y) = \delta_0 + \delta_1 y$. This embeds a “slippery” job ladder (Jarosch (2023)) into our model. Second, we allow there to be exogenous job mobility shocks (“godfather shocks”), λ_g , under which a worker draws a job offer and either can accept the offer or receive the value of unemployment. Finally, we introduce moving costs η_e for EE transitions and η_u for UE transitions. Below we outline the value functions, dropping the state variables (z, p) as in Section 2. The equivalent result to Proposition 1 in this environment follows from our proportionality assumption on moving costs.

Moving costs. Moving costs represent the utility loss workers experience when transitioning between jobs or from unemployment to employment.²² We assume these costs are proportional to the workers' net value of the new job:

$$\eta_e(y) = \eta_e \cdot [W(\hat{w}, y, \varepsilon', g'_\varepsilon) - U] \quad (12)$$

$$\eta_u(y) = \eta_u \cdot [W(\hat{w}, y, \varepsilon', g'_\varepsilon) - U] \quad (13)$$

where η_e and η_u are proportionality constants for job-to-job and unemployment-to-job transitions respectively, and $\hat{w}$ is the wage such that the worker receives the full value of the match such that $J(\hat{w}, y, \varepsilon', g'_\varepsilon) = 0$.

Moving costs enter the model through the wage determination process. When a worker at firm y meets firm $x > \hat{y}(y)$ (poaching), the wage is set such that:

$$W(\phi^{\text{poach}}(x, y, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_e(x) = W(w', y, \varepsilon', g'_\varepsilon) \quad (14)$$

²²We assume that workers do not face moving costs when laid-off.

When a worker at firm y meets firm $x \in [q(w', y, \varepsilon', g'_\varepsilon), \hat{y}(y)]$ (renegotiation), the wage is set such that:

$$W(\phi^{\text{reneg}}(y, x, \varepsilon', g'_\varepsilon), y, \varepsilon', g'_\varepsilon) = W(\hat{w}(x, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_e(x) \quad (15)$$

Finally, when a worker who is unemployed meets firm x , the wage is now set such that:

$$W(\phi_u(x, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_u(x) = U \quad (16)$$

These conditions ensure that after accounting for moving costs, workers are indifferent between their options. The moving cost structure implies that transitions to higher surplus jobs require larger wage premia to compensate workers for the utility loss of moving.

Worker and Firm values. Including these moving costs, and additionally accounting for the godfather shocks and the slippery job ladder, the value of employment to a worker is:

$$\begin{aligned} W(w, y, \varepsilon, g_\varepsilon) = & \max_{s \in [0, \bar{s}_e]} \frac{w}{\varepsilon} - c(s) + \beta(\delta(y) + (1 - \delta(y))\lambda_g)U \\ & + \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta(y)) (1 - \lambda_g) (s + \lambda_e) \int_{\underline{y}}^{q(w', y, \varepsilon', g'_\varepsilon)} W(w', y, \varepsilon', g'_\varepsilon) dF(x) \\ & + \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta(y)) (1 - \lambda_g) (s + \lambda_e) \int_{q(w', y, \varepsilon', g'_\varepsilon)}^{\hat{y}(y)} W(\phi^{\text{reneg}}(y, x, \varepsilon', g'_\varepsilon), y, \varepsilon', g'_\varepsilon) dF(x) \\ & + \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta(y)) (1 - \lambda_g) (s + \lambda_e) \int_{\hat{y}(y)}^{\bar{y}} [W(\phi^{\text{poach}}(x, y, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_e(x)] dF(x) \\ & + \beta \mathbf{E}_{g'_\varepsilon} (1 - \delta(y)) (1 - \lambda_g) (1 - s - \lambda_e) W(w', y, \varepsilon', g'_\varepsilon) \end{aligned}$$

where $\varepsilon' = \varepsilon(1 + g'_\varepsilon)$. Note also that because of the non-zero moving costs, the cutoff firm determining mobility is defined as $\hat{y}(y)$.

The value of unemployment is:

$$\begin{aligned} U = & \max_{s \in [0, \bar{s}_u]} b - c(s) + \beta(1 - (s + \lambda_u))U \\ & + \beta(\lambda_u + s) \mathbf{E}_{g'_\varepsilon} \int_{\underline{y}}^{\bar{y}} \max\{W(\phi_u(x, \varepsilon', g'_\varepsilon), x, \varepsilon', g'_\varepsilon) - \eta_u(x), U\} dF(x) \end{aligned} \quad (17)$$

As before, we have the result that:

$$U = \frac{b}{1 - \beta} \quad (18)$$

due to the “take it or leave it” offers.

The value to the firm is:

$$\begin{aligned}
J(w, y, \varepsilon, g_\varepsilon) &= y - \frac{w}{\varepsilon} \\
&+ \beta(1 - \delta(y))(1 - \lambda_g) (s_e^*(w, y, \varepsilon, g_\varepsilon) + \lambda_e) \mathbf{E}_{g'_\varepsilon} \int_{q(w', y, \varepsilon', g'_\varepsilon)}^{\hat{y}(y)} J(\phi^{\text{reneg}}, y, \varepsilon', g'_\varepsilon) dF(x) \\
&+ \beta(1 - \delta(y))(1 - \lambda_g) (s_e^*(w, y, \varepsilon, g_\varepsilon) + \lambda_e) \mathbf{E}_{g'_\varepsilon} \left[\int^{q(w', y, \varepsilon', g'_\varepsilon)} dF(x) \right] J(w', y, \varepsilon', g'_\varepsilon) \\
&+ \beta(1 - \delta(y))(1 - \lambda_g) \left[(1 - s_e^*(w, y, \varepsilon, g_\varepsilon) - \lambda_e) \right] \mathbf{E}_{g'_\varepsilon} J(w', y, \varepsilon', g'_\varepsilon)
\end{aligned}$$

where $s_e^*(\cdot)$ is the optimal search effort of the worker.

E Model Solution and Simulation

The value and policy functions are solved in the following steps:

1. For a given home production function b , output functions y , discount rate β , exogenous separation probability δ (or $\delta(y)$ in the extended model), the distribution of vacancies $F(y)$, the arrival probabilities λ_u , λ_e (and λ_g , η_e , η_u for the extended model), and a fixed guess for the search policy functions, solve for U , $W(\cdot)$, and $J(\cdot)$.
2. Update the search policy functions given the converged values.
3. Repeat steps 1-2 until convergence.

To simulate data from our model for the COVID-like ε shock we take the following steps:

1. We draw 4000 paths for g_ε over 4000 periods. We place a restriction on the paths so that in period $\tau - 1$, the grid for g_ε is set so that $g_\varepsilon = 0$, and so that $g_\varepsilon = 0.0058$ in period τ , simulating dynamics from this point forward.²³
2. We simulate labor market histories for 20000 workers in each of the 4000 economies. In period 1, all workers are unemployed.
3. For each period t from 2 to 4000, the simulation evolves the labor market state for all workers. In each period, the simulation: (i) determines the current inflation state (setting $g_\varepsilon = 0$ before period τ , $g_\varepsilon = 0.0058$ at period τ , and allowing g_ε to evolve stochastically according to the Markov transition matrix thereafter); (ii) draws firm meeting opportunities for both unemployed workers (with probability λ_u) and employed workers (with

²³We set period $\tau = 432$ and note that on an annualized basis, $g_\varepsilon + g_p \approx 0.09$, roughly equivalent to the

probability $\lambda_e + s(w, y, p, \varepsilon, g_\varepsilon)$, where search intensity depends on the worker's current wage, firm productivity, and the price level); (iii) determines exogenous job separations based on the separation probability $\delta(y)$; (iv) draws potential new firm matches from the productivity distribution; and (v) updates each worker's employment status, firm match, wage, and contract state based on optimal acceptance and mobility decisions given the value functions.

4. After simulating the full labor market dynamics, we compute welfare measures for each worker in each period from τ forward. For each worker-period, we calculate flow values including wages net of search effort costs for employed workers, home production value b for unemployed workers, and mobility costs incurred when switching firms. These flow values are then discounted back to period τ using the model's discount factor β .
5. We aggregate welfare measures across workers and time periods. We focus on workers who were employed at the time of the shock (period τ) and compute the present discounted value of their welfare, wages, search costs, and mobility costs over the entire post-shock period. We also compute overall match surplus measures that include both worker and firm values. These statistics are averaged across all 4000 simulated economies to obtain expected welfare effects of the inflation shock. We also trace outcomes of interest in the periods following the shock, including average search effort, the job-to-job transition rate, average real wages, renegotiation rates.

To simulate the stochastic steady state for different inflation volatility regimes we take the following steps:

1. We construct discrete price grids and price transition matrices using the Rouwenhorst method, keeping ρ fixed at our estimated value for the US but varying volatility σ_ν over a large grid indexed by q . Each scenario is characterized by its own Markov transition matrix P_ε governing the evolution of the cyclical inflation component g_ε .
2. For each volatility scenario q , we solve for the model's equilibrium value functions given the price grid and transition matrix. We compute both the social match value $\tilde{M}(w, y, p, \varepsilon, g_\varepsilon)$ and social unemployment value $\tilde{U}(p, \varepsilon, g_\varepsilon)$ defined in Appendix C, as well as the worker value functions $W(\cdot)$ and U . These value functions incorporate workers' privately optimal search decisions $s_e^*(w, y, p, \varepsilon, g_\varepsilon)$ under the specified inflation process.
3. For each volatility scenario q , we run $N = 3600$ independent Monte Carlo simulations of the economy's stochastic steady state. In each simulation, we simulate labor market histories for workers over T periods, where T is sufficiently large to ensure convergence to steady state. In period 1, all workers begin unemployed.

4. For each period t from 2 to T , the simulation evolves the labor market state for all workers. In each period, the simulation: (i) draws the current inflation state ε_t according to the Markov transition matrix P_ε ; (ii) draws firm meeting opportunities for unemployed workers (with arrival rate λ_u) and employed workers (with endogenous arrival rate $\lambda_e + s^*(w, y, p)$, where search intensity depends on the worker's current wage, firm productivity, and price level); (iii) determines exogenous job separations based on firm-specific separation rates $\delta(y)$; (iv) draws potential new firm matches from the productivity distribution $F(y)$; (v) evaluates workers' continuation wages accounting for inflation realizations and potential renegotiation; and (vi) updates each worker's employment status, firm match, wage, and contract state based on optimal acceptance and mobility decisions given the value functions.
5. After the labor market reaches stochastic steady state (defined as the last 1000 periods of the simulation), we compute welfare measures. For each worker in each steady-state period, we evaluate their social match value $\tilde{M}(\cdot)$ if employed or social unemployment value $\tilde{U}(\cdot)$ if unemployed, their worker values $W(\cdot)$ or U , flow search costs $C(s^*, z)$, and mobility costs $\eta_e(y, z)$ or $\eta_u(y, z)$ when switching employers. We calculate both employment-weighted average welfare measures and welfare statistics accordingly.

F Cyclical Movements in the Arrival Rate of Offers

In this section, we extend the model to allow for the arrival rate of offers to move with the cyclical rate of inflation. If the exogenous component of the arrival rate co-moves positively with inflation shocks, a higher offer arrival rate might mitigate some of the costs associated with an inflation shock. To allow for this possibility, we assume that:

$$\lambda_u \equiv \bar{\lambda}_u + \psi_u g_\varepsilon$$

$$\lambda_e \equiv \bar{\lambda}_e + \psi_e g_\varepsilon$$

We assume a common factor $\psi = \psi_e = \psi_u$. We discipline ψ using the relationship between the *cyclical* component of the job finding rate (which is determined solely by the exogenous component of the arrival rate since search effort is zero for the unemployed) and the detrended inflation rate, finding a slope coefficient of 6.²⁴ Accordingly, we set $\psi = 6$ in our quantitative exercise.

maximum value of the 12-month CPI inflation rate in 2022.

²⁴Formally, we consider the regression $UE_t = \alpha + \psi \pi_t + r_t$, where π_t is the detrended 12-month inflation rate, converted to monthly frequency, and UE_t is the monthly job finding rate.

Exactly as done in Section 4.1, we use our extended model to consider an inflationary episode similar in magnitude to the COVID-19 pandemic. In the extended model, the large persistent shock to g_ε is also a shock to λ_e , elevating the *passive* offer arrival rate from 0.0467 to 0.0824 upon impact, before gradually settling down, moving in tandem with g_ε .

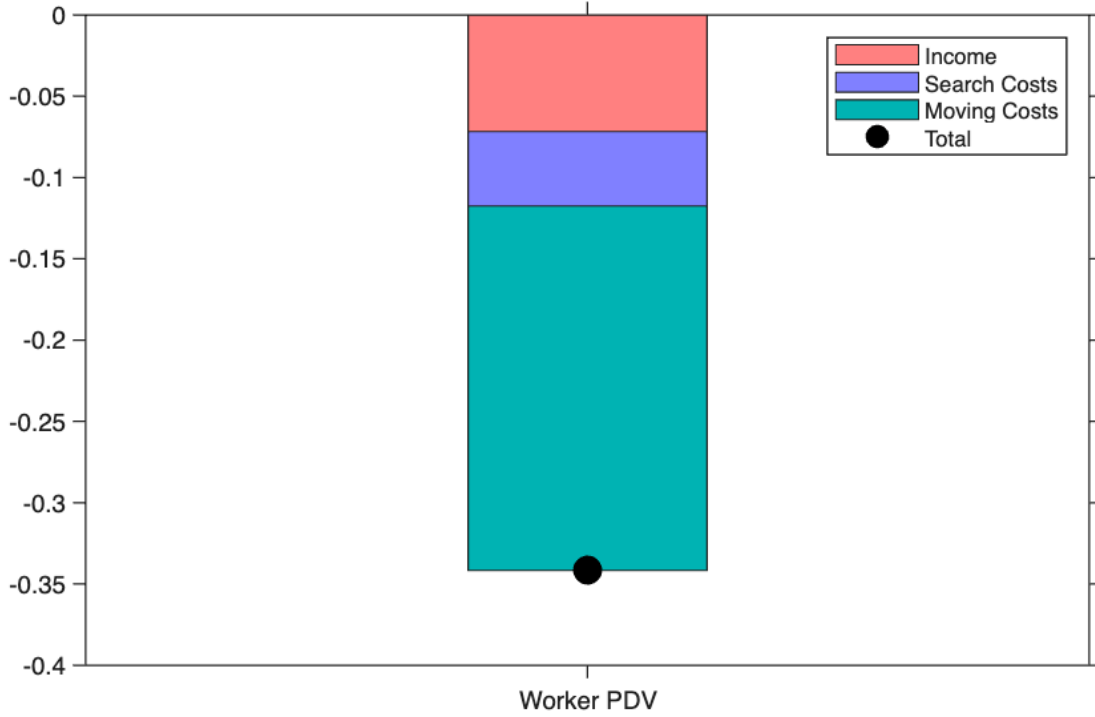


Figure 10: Lifetime Worker Welfare Costs: Cyclical Arrival-Rate Extension

Note: Figure 10 plots the mean percent difference in the present discounted value of worker income net of search and moving costs in the shocked economy and the baseline one as a black dot. The colored bars decompose this difference in PDVs into a component explained by the difference in PDV of income, a component explained by the difference in the PDV of search costs, and a component explained by the difference in the PDV of moving costs.

Our first finding (Figure 10) is that allowing for a substantial deterministic relationship between g_ε and λ_e has only a marginal effect on the welfare losses experienced by workers: on average, workers are around 0.34% worse off as a result of the shock in the economy where $\psi = 6$, whereas they are 0.44% worse off in the economy where $\psi = 0$. However, the underlying drivers of the losses are quite different, with only 21% of the loss coming from income losses, 13% from search costs and 66% from moving costs. As λ_e enters the worker's contact rate additively, movements in the passive offer arrival rate only influence search effort indirectly through an increase in the speed of wage adjustment and movement up the job ladder. Therefore, allowing for a deterministic relationship between λ_e and g_ε only somewhat reduces search costs borne by

workers, while substantially attenuating income losses. On the other hand, the increase in λ_e increases moving costs borne by workers because it increases the job-to-job transition rate.

Unsurprisingly, allowing for a substantial deterministic relationship between g_e and λ_e has a similar effect on *social* welfare changes resulting from the large inflationary shock, which flip from a small loss to a gain of roughly 0.13 percent of baseline (Figure 11). The main channel for this net change is that the increase in λ_e results in more workers being allocated up the job ladder, elevating output dramatically, an effect that is only somewhat offset by the increase in moving costs.

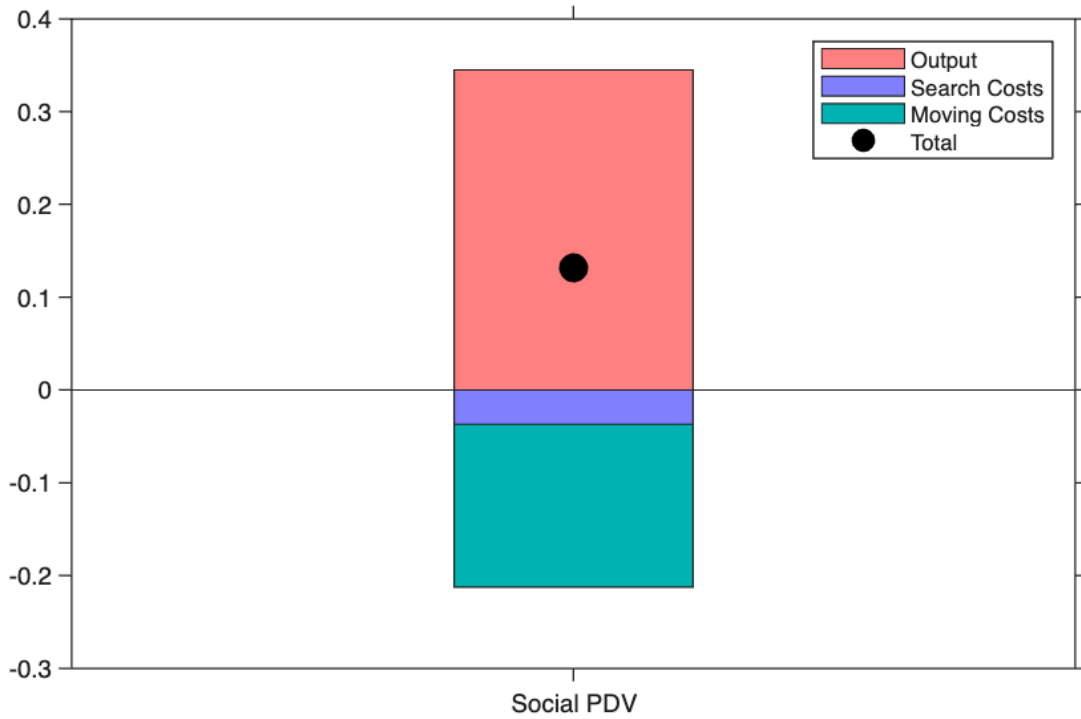


Figure 11: Welfare Costs of an Inflationary Shock: Cyclical Arrival-Rate Extension

Note: Figure 11 plots the mean percent difference in the present discounted value of output net of search and moving costs in the shocked economy and the baseline one as a black dot. The colored bars decompose this difference in PDVs into a component explained by the difference in PDV of output, a component explained by the difference in the PDV of search costs, and a component explained by the difference in the PDV of moving costs.